

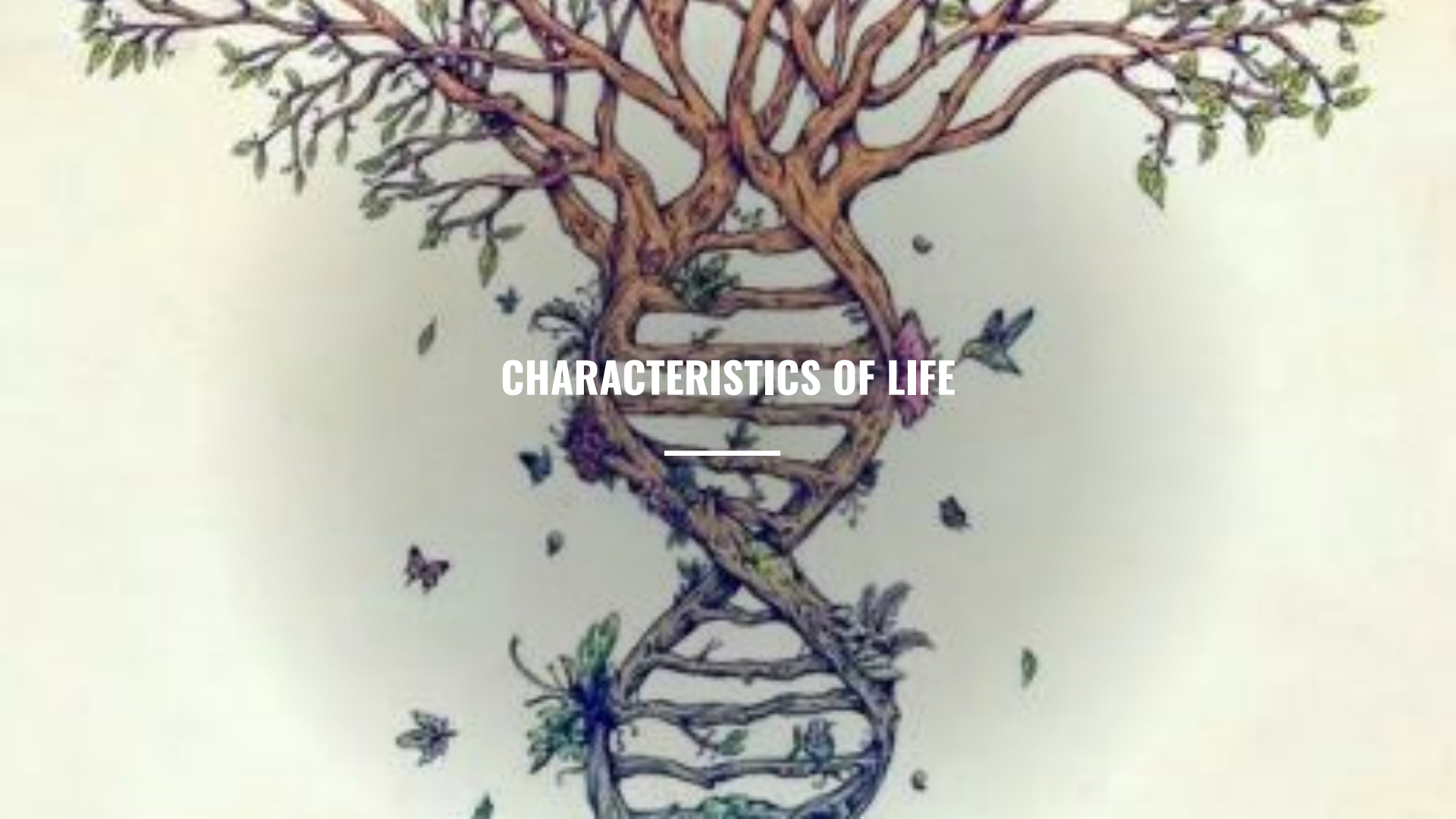
UNIT 1- BIOCHEMISTRY

-Characteristics of Life

-Graphing, Measurements, & Experimental Design

-Properties of Water

-Types of Reactions & Biomolecules



CHARACTERISTICS OF LIFE

Organization of Life

- **Matter** = anything that has mass and occupies space

- Composed of **atoms**

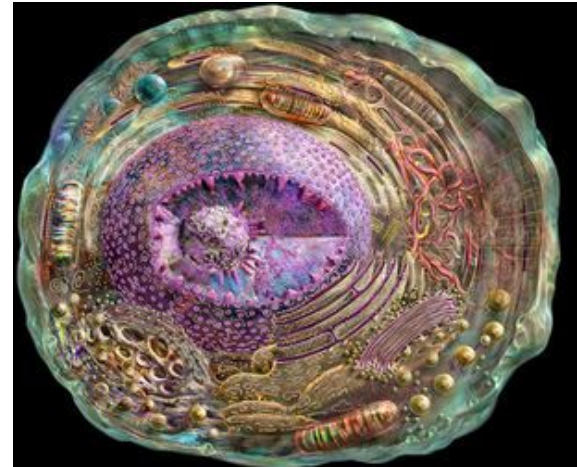
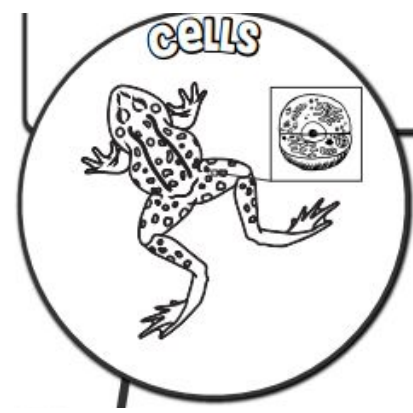
- The smallest part of an element (*think PT*) that still has the properties of that element

- 6 elements necessary for life = **C, H, O, N, P, S**
- **Cell** = most basic unit of life



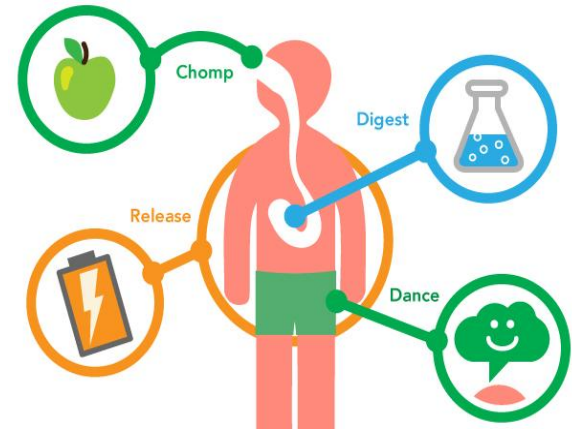
1. CELLULAR ORGANIZATION

- All living things are **made of cells**
- Can be multicellular (many cells) or unicellular (one cell) and can live on their own or in colonies (groups)
- All cells have the same basic structure



2. Energy Utilization

- **Metabolism** - the sum of ALL chemical reactions that happen in an organism- **Obtaining and using energy**
- These reactions include:
 - Making energy
 - Autotroph: an organism that produces its own food
 - Heterotroph: an organism that needs to consume other organisms
 - Making new tissues
 - Destroying pathogens (invaders)



3. GROWTH AND DEVELOPMENT

- All living organisms grow!
- **Growth**- increase in the amount of living material and formation of new structures
- **Development**- changes that take place during the life of an organism



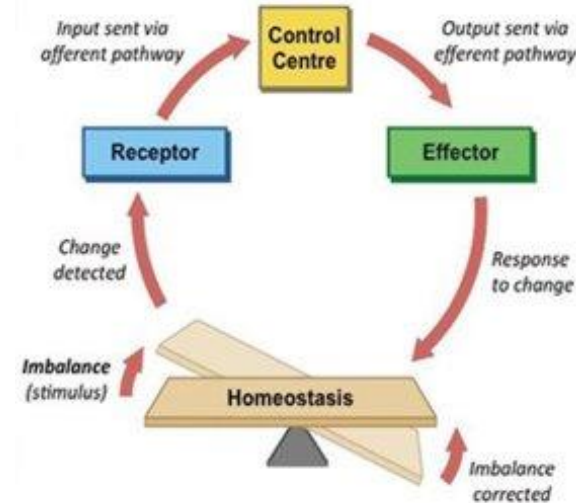
4. REPRODUCTION AND HEREDITY

- Goal of reproduction: **pass on genetic material to offspring**
- Reproduction is essential to survival of a species (but not individuals)
- **Genes** are passed on in the form of **DNA** from parents to offspring either by sexual reproduction (with 2 parents) or asexual reproduction (with 1 parent)
 - **Genes determine characteristics of the offspring**



5. INTERNAL REGULATION (HOMEOSTASIS)

- **Homeostasis** - maintaining a **stable, internal condition** despite changes in external environment
 - Inside cells: Water/salt concentrations, waste, etc.
 - Inside organisms: body temperature, blood sugar, etc.
- Internal regulation: organisms have **optimal ranges** for best function (temperature, pH, water regulation, etc.)
 - Homeostasis keeps organisms in that range



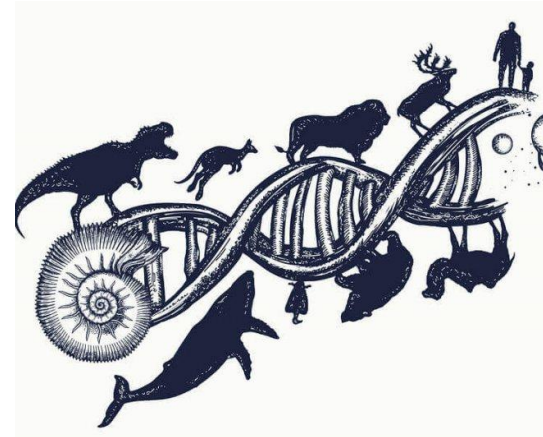
6. RESPONSE TO ENVIRONMENT

- All organisms respond to their environment in order to survive
- Stimulus- a factor that requires an organism to adjust
- Response- the means by which the organism reacts to the stimulus



7. EVOLUTION AND ADAPTATION

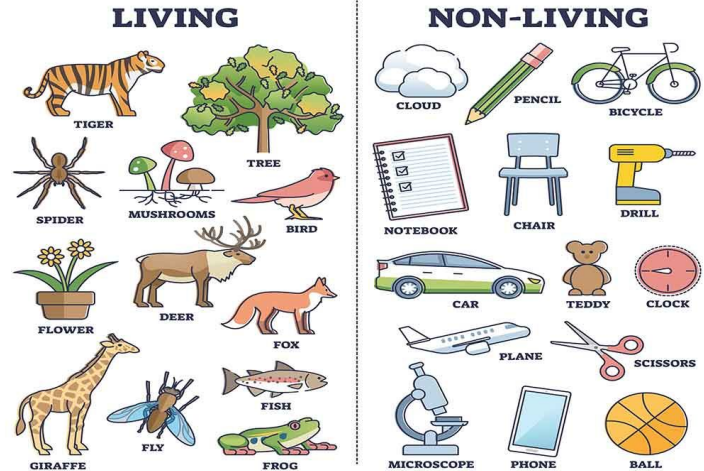
- Evolution- changes in characteristics of a population over **multiple generations**
- A population will *gradually* change over time in response to environmental conditions
- Adaptation- traits that allow the organism to be best fit in their environment
 - Only the *best-suited* organisms will survive and reproduce!



LIVING OR NOT?

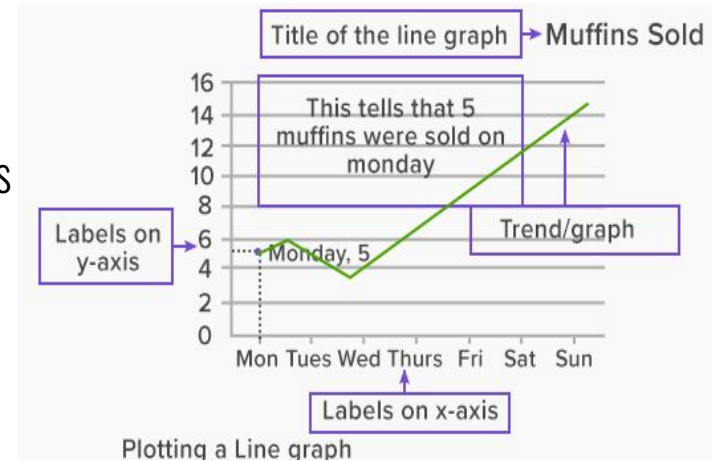
Central idea: In order to be classified as a living thing, something must exhibit ALL characteristics of living things. Not 5, not 6, ALL of them.

In the bottom left corner of your Characteristics of Life notes, try to classify viruses as living or nonliving based on what you know and/or have experienced.



WHAT IS A GRAPH?

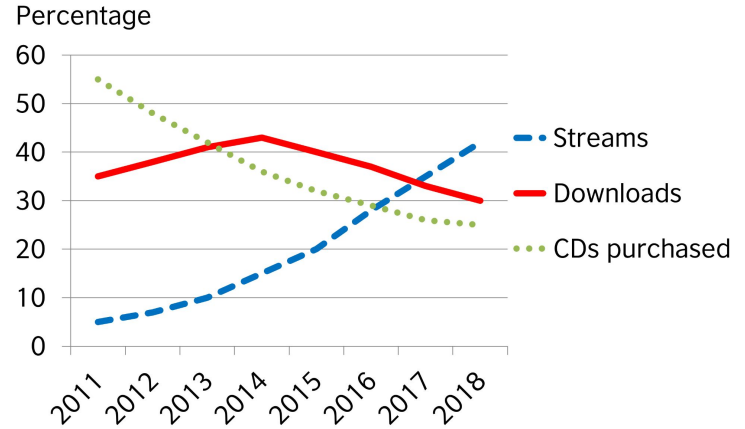
- **X-axis** – horizontal axis □ **independent variable** (“I” **manipulated** this, e.g. time, amount of medicine given, etc.)
- **Y-axis** – vertical axis □ **dependent variable** (I **measured** this during the experiment, e.g. amount of gas produced, change in height, etc.)
- **All axes must have titles, labels, and units**
- Proper scale is important to show accurate patterns & trends!



TYPES OF GRAPHS-LINE GRAPHS

- Used to show the relationship between an independent and dependent variable
 - Often time will be the independent variable on X-axis
 - **ALWAYS** use line graphs when measuring change over time
- Can compare more than one event *if the scale is the same*
- Most commonly used graph in science

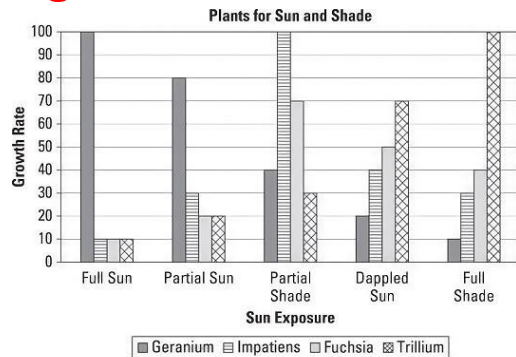
Percentage of total music sales by method



TYPES OF GRAPH: BAR AND PIE

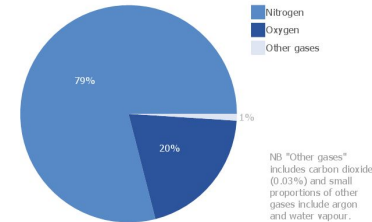
BAR GRAPH

- Used to compare a **measured amount between groups or categories**



PIE GRAPH

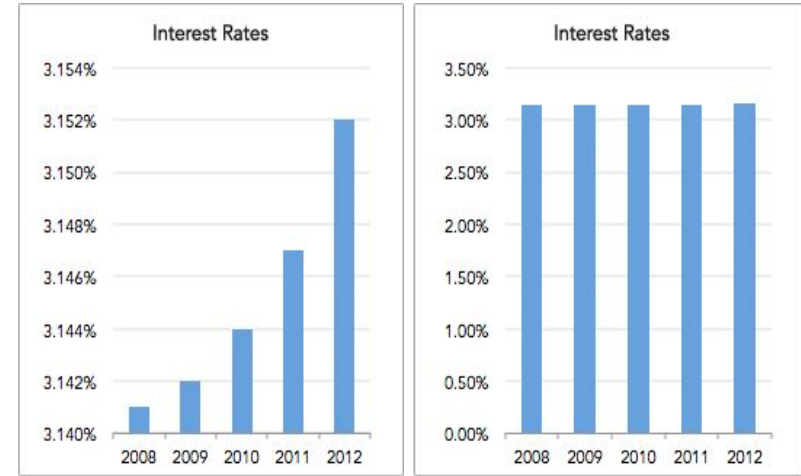
- Used to show **percentages** or how something can be broken down into **parts**



ISSUES WITH GRAPHS

- Using the wrong scale can skew data (can be intentional or unintentional)
 - Altering the scale can be used to distort data (intentional or not)
- Using more than one unit will result in an inaccurate graph

Same Data, Different Y-Axis



GRAPHING QUICK TIPS

- Use the range of the data to determine scale
- If the scale unit is too large, your graph will be cramped into a small area and will be hard to read and interpret
- If the scale unit is too small, the graph will run off the paper
- **Multiples of 1, 2, 5, or 10 are easiest to work with.**
- When choosing between a bar and a line graph:

Use **line graphs** for **change over time** or to **show a relationship** between two variables

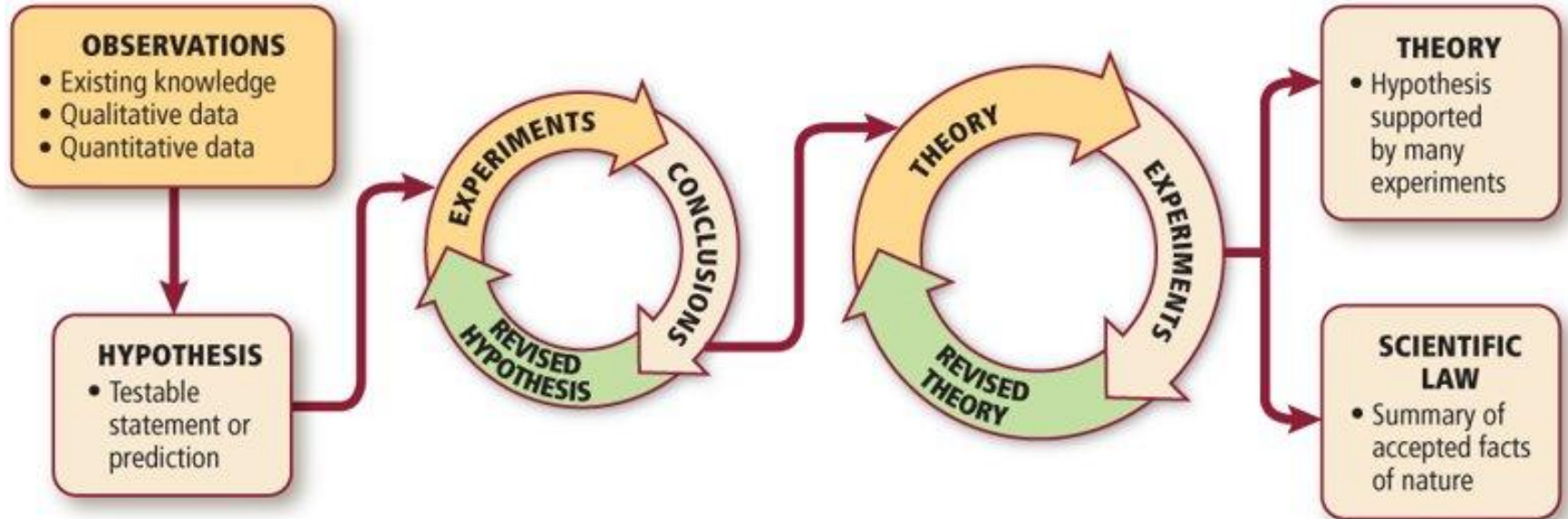
Use **bar graphs** when **comparing different groups**

EXPERIMENTAL
DESIGN

The Scientific Method

- **Scientific method** - systematic approach used in scientific study, whether it is chemistry, physics, biology, or another science.
- Science is based on **experimentation**- our understanding is always being built upon with new findings

THE PROCESS



TYPES OF DATA

QUALITATIVE

- Describes **qualities**
- Examples:
 - Green color
 - Large size
 - Sour Taste
 - Sweet Smell

QUANTITATIVE

- Uses **numerical values** to measure something
- Examples:
 - 4 feet long
 - 6 legs
 - 7.2 grams
 - 100 mL

What kind of
qualitative
observations can
we make?
What about
quantitative?



ASKING QUESTIONS

OBSERVATIONS

- Description of something you can see, smell, taste, touch, or hear
- NOT an opinion (it is **objective**)
- Example:
 - The ground is wet.

INFERENCES

- A guess about a situation or outcome *based on observations*
- You can make **many inferences** from a **single observation**
- Example:
 - The ground is wet because it rained.
 - The ground is wet because someone was watering their plants.



What are some observations we can make?

What inferences can we make based on those observations?



Nature of Science

Key scientific terms

- Law: broad concept or principle (**HOW**)
- **Describes** patterns in nature
- Often accepted as facts
- Examples:
 - Newton's laws of motion
 - Boyle's gas laws
 - Law of Conservation of Mass
- Theory: An **explanation** of an observed phenomenon
- Organizes facts and research from scientists to explain **WHY**
- NEVER becomes a fact or a law
- Example:
 - Evolutionary theory

Construct a hypothesis.

What do you think might happen?

- **Hypothesis**: A testable prediction based on observations (*more than a guess!!*) that describes a cause and effect relationship between variables

Format for a Hypothesis:

“If (IV) then (DV). “

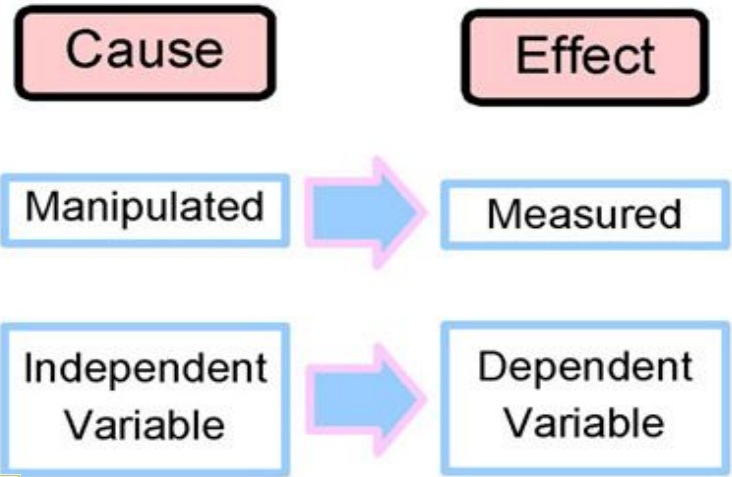
IV = Independent variable = Cause

DV = Dependent variable = Effect

Types of Variables

- **Independent Variable**: (IV) what the experimenter **deliberately changes or manipulates** in the investigation
 - Usually on the **X-Axis** of a graph (horizontal)
 - Should be the ONLY thing different between experimental groups
 - Ex. You want to do an experiment to see if what you drink before a race affects how quickly you run. *What is the IV?*
- **Dependent Variable**: (DV) what **changes/occurs in response** to the independent variable
 - On the **Y-Axis** of a graph (vertical)
 - Usually represented by the data you collect; what is **measured**
 - Ex. You want to do an experiment to see if what you drink before a race affects how quickly you run. *What is the DV?*

INDEPENDENT VARIABLE VS. DEPENDENT VARIABLE



Independent variable

The variable the researcher changes.

Application of fertilizer 'x' in this experiment

vs

Dependent variable

Variable affected by change in independent variable

- Plant growth
- No. of leaves,
- No. of fruits etc in this experiment.



EXPERIMENTAL vs. CONTROL GROUP

- **experimental group**- receives the **treatment** or is **manipulated** in some way
 - exposed to the independent variable; can have multiple in an experiment
- **control group**- “standard group” that does not receive treatment; **used for comparison**
 - If there is a difference between the groups, having a control group(s) increases the likelihood (but doesn’t “prove”!) that the independent variable actually caused the difference.



CONSTANTS

- Between all experimental groups, **the only thing that should change is the independent variable.**
- Constants- aspects of an experiment that are held **consistent or the same**
 - Ensures that all aspects of trials are identical, except for the IV
 - Ensures that any difference measured in the DV is caused only by the IV
 - *Ex. All runners in an experiment should be the same age, gender, same breakfast, same training, same shoes, etc.*
- Having as many **repeated trials** as possible to:
 - Ensure the results aren't due to chance
 - Eliminate any errors
 - Ensure the data is precise

Experimental Design Example

- I want to run an experiment testing **how caffeine affects the ability of grizzly bears to solve puzzles.**
- There are **four groups** of bears. Group A gets no caffeine, Group B gets 100 mg of caffeine, Group C gets 200 mg of caffeine, and group D gets 400 mg of caffeine
- The bears then all solve the **same puzzle** and then time it takes them to solve it is measured.

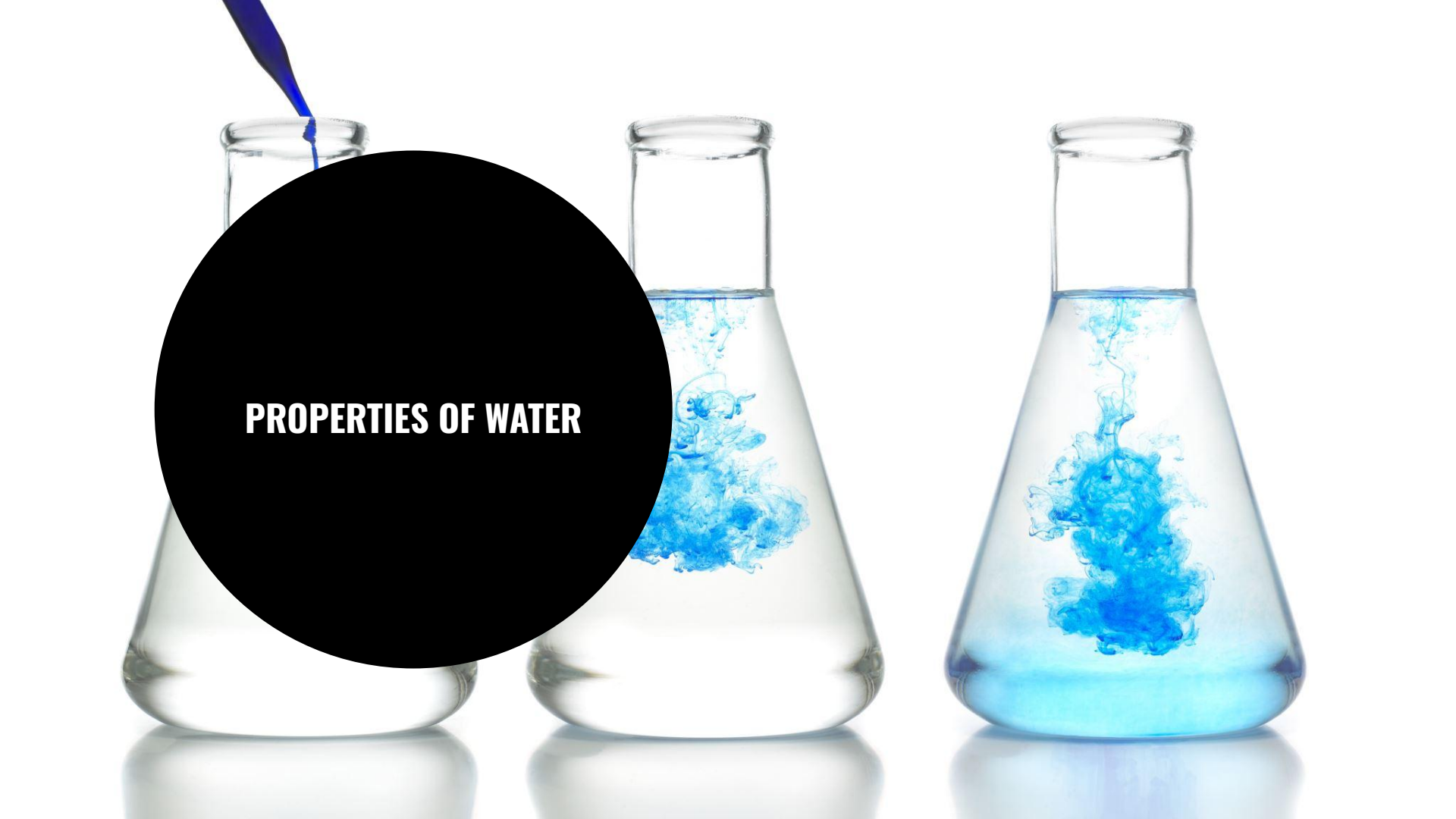


- Control Group:
- Experimental Group:
- Independent Variable:
- Dependent Variable:
- Constants:

Experimental Design Example



- I want to run an experiment testing **how caffeine affects the ability of grizzly bears to solve puzzles.**
- There are **four groups** of bears. Group A gets no caffeine, Group B gets 100 mg of caffeine, Group C gets 200 mg of caffeine, and group D gets 400 mg of caffeine
- The bears then all solve the **same puzzle** and then time it takes them to solve it is measured.
- Control Group: **Group A (received no treatment)**
- Experimental Group: **Groups B, C, and D (received the treatment)**
- Independent Variable: **Amount of caffeine (0, 100, 200, 400 mg - what they changed)**
- Dependent Variable: **Time it took for the bears to solve the puzzle (what they measured)**
- Constants: **Same type of bear (grizzly), same puzzle, assume same delivery of caffeine (e.g. pill, injection, etc.), same number of bears in each group....**

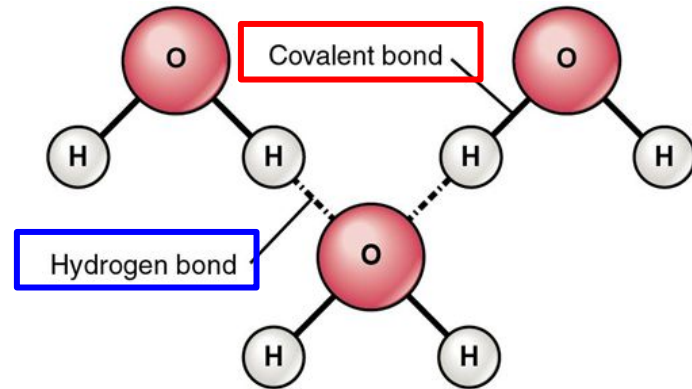
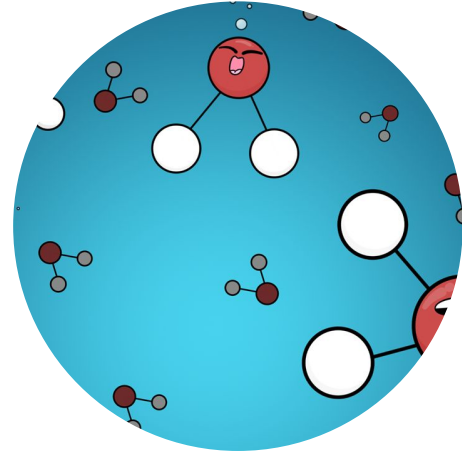


The image shows three Erlenmeyer flasks arranged horizontally on a white reflective surface. The leftmost flask is being filled with a blue liquid from a pipette. The middle flask shows the blue liquid just beginning to disperse from the surface. The rightmost flask shows the blue liquid fully mixed throughout the water. A large black circle is superimposed over the left and middle flasks, containing the text 'PROPERTIES OF WATER'.

PROPERTIES OF WATER

WATER- THE MOLECULE OF LIFE

- H and O atoms in **each H_2O molecule** are held together by **covalent bonds** (formed when two atoms **SHARE** electron pairs)
- **Separate H_2O molecules** are attracted together by **hydrogen bonds**
- These relatively strong bonds allow water to have many properties essential to life



HYDROPHILIC VS. HYDROPHOBIC

- **Hydrophilic** substances- Water “loving” → have an affinity for water

- Usually dissolve easily in water
- typically **ions or polar molecules**
- Example: **salt and sugar**



- **Hydrophobic** substances- water “fearing” → do **not** have an affinity for water

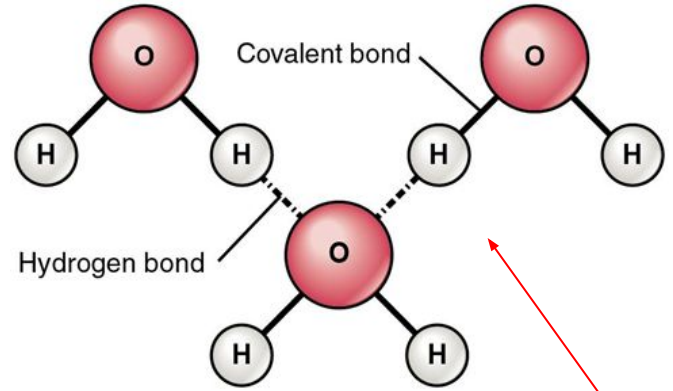
- Does not dissolve in water
- typically **nonpolar** substances
- Example: **oil**



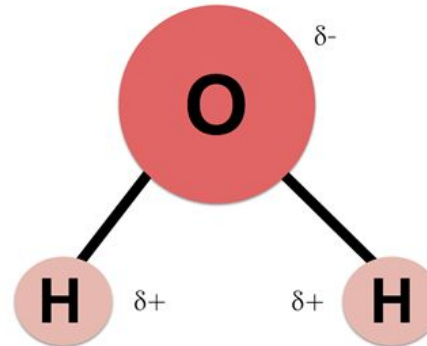
Figure 1- Image of water droplet on lotus leaf

WATER IS POLAR

- Water is a **polar** molecule, meaning **electrons are not shared equally**
- Oxygen attracts electrons more strongly than hydrogen
- Leads to **asymmetrical charge**: H side is partially positive and O side is partially negative
- **Hydrogen bonds are weaker** than covalent or ionic bonds

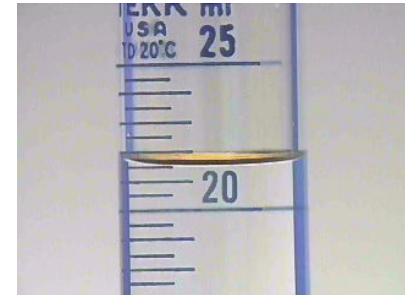
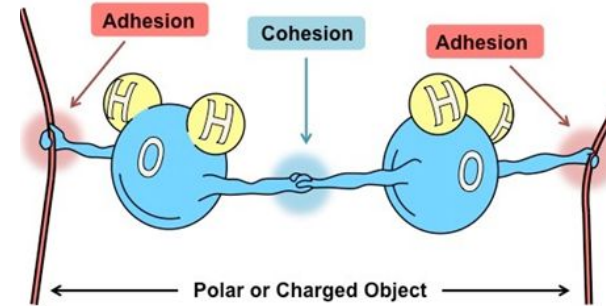


Hydrogen bonds form **BETWEEN** water molecules and covalent bonds form **WITHIN** one water molecule.



PROPERTY 1: CAPILLARY ACTION

- **capillary action**- water can move **upwards** due to strong cohesive and adhesive properties
 - Ex. water moves up thin plant vessels or drinking straws
- **Cohesion**- (“co-” prefix = with or together) attraction of molecules of the **same** substance
 - Ex. water sticking to water molecules; allows round droplets to form on leaves and causes **high surface tension**
- **Adhesion**- attraction of molecules of **different** substances
 - Ex. water molecules sticking to the glass on graduated cylinders causes a **meniscus** to form



Meniscus (FYI: Read the **bottom** of the curve for accurate volume measurement.)

PROPERTY 2: SURFACE TENSION

- **Surface tension**- a measure of how difficult it is to stretch or break the **surface** of a liquid
- **Water** has a **high surface tension** due to its **cohesive** properties (caused by **hydrogen bonds**)



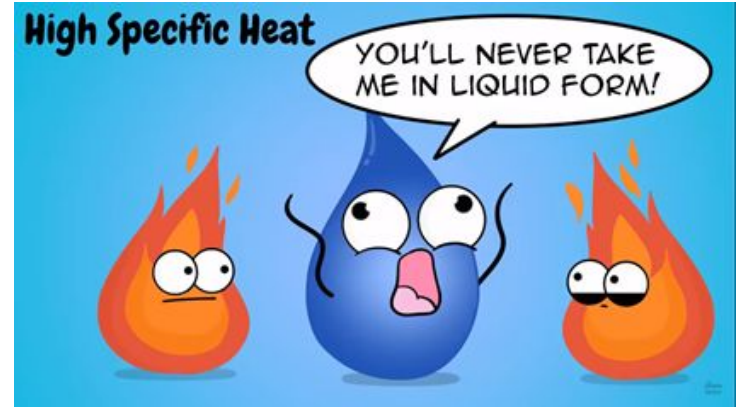
PROPERTY 3: HIGH SPECIFIC HEAT

- **Specific heat**- the amount of energy required to raise the temperature of 1 gram of a substance by 1°C
- **Water** has a relatively **high specific heat** due to **hydrogen bonding** between water molecules

In other words, the temperature of water does not change as easily as other substances.

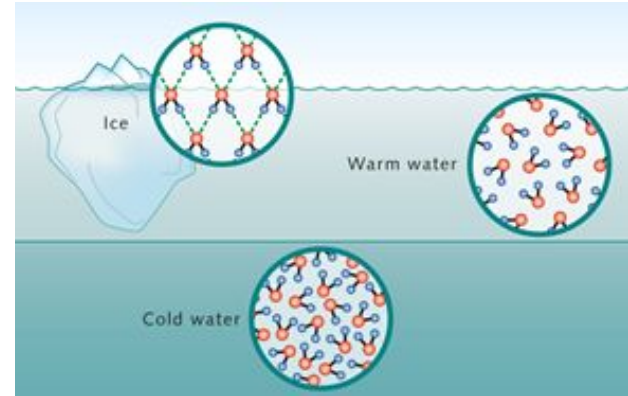
- keeps environments stable and **allows for life in water**

Ex. A pond temperature will stay relatively constant from day to night



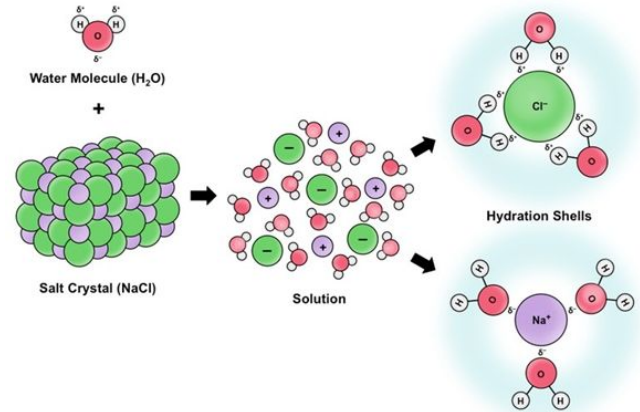
PROPERTY 4: LESS DENSE AS A SOLID

- Water is **less dense** as a **solid** than it is a liquid (water expands as it freezes)
- Causes ice to float and prevents aquatic environments from freezing solid
- Ex. Life can exist under frozen surfaces of lakes or oceans



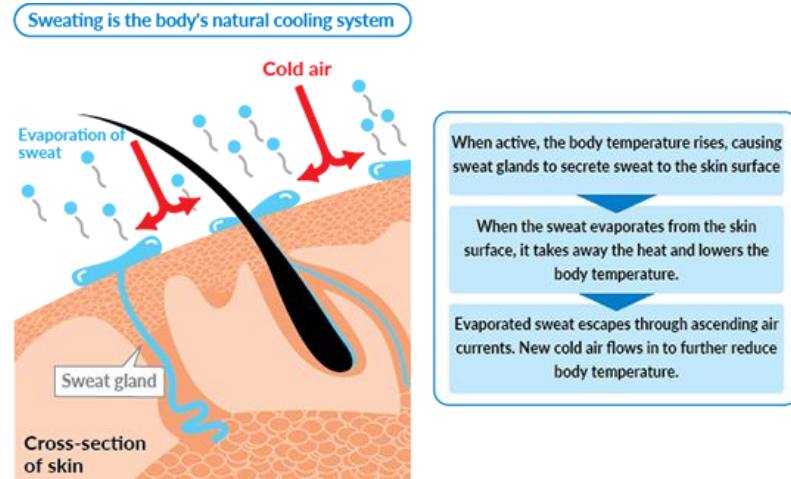
PROPERTY 5- THE UNIVERSAL SOLVENT

- **Solution**- a liquid that is a mixture of two or more substances (e.g. sugar water)
- **Solvent**- the dissolving agent (e.g. water)
- **Solute**- the substance that is dissolved (e.g. sugar or salt)
- **Water** is known as a “universal solvent” because its **polarity (unequal charge distribution)** allows many different substances like charged ions to be dissolved easily



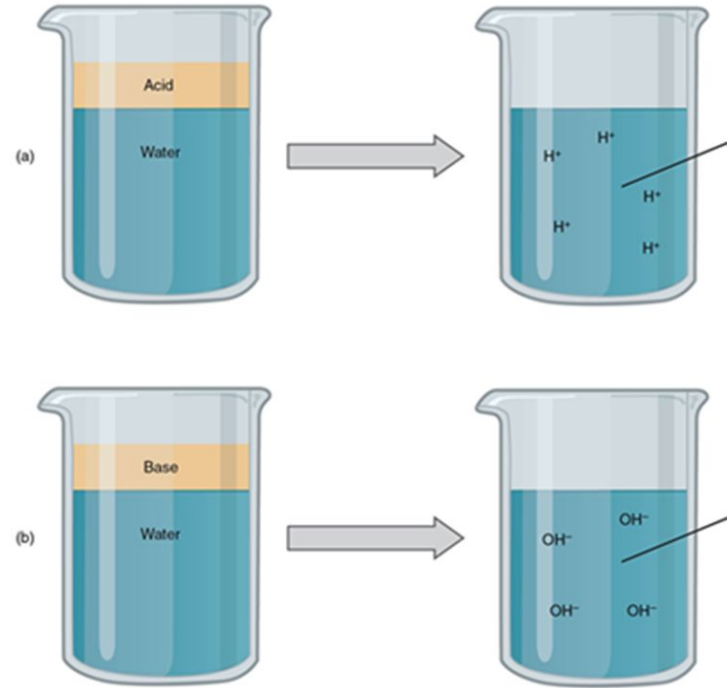
PROPERTY 6- HIGH HEAT OF VAPORIZATION

- **Heat of vaporization**- the amount of energy required to turn a liquid to a gas
- **Evaporative cooling**- as liquid evaporates, the surface of the liquid left behind cools down, making it harder to continue the evaporation process
- Water draws a high amount of energy from your skin to evaporate → **why sweating helps keep your body cool**
- Also helps water on Earth and your body to stay in liquid form!



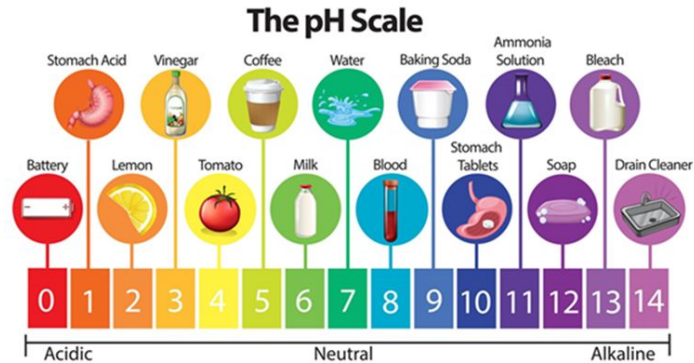
pH

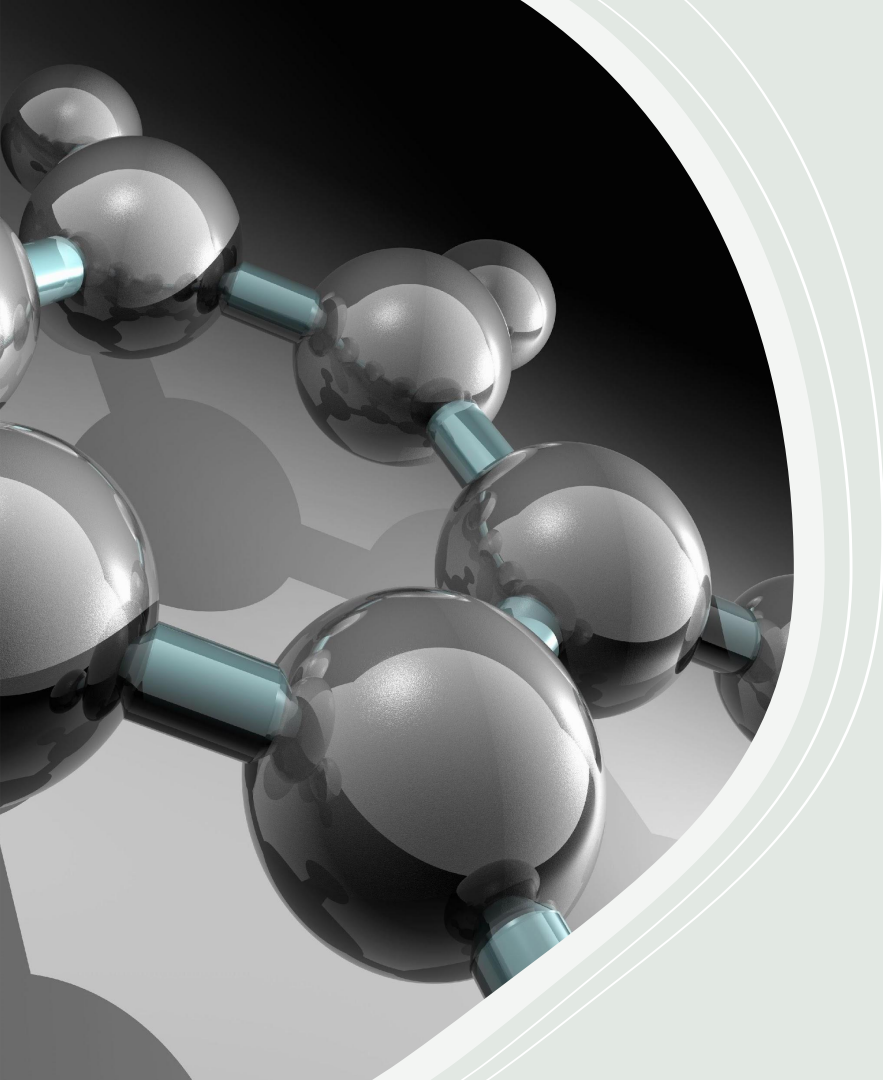
- Water **dissociates** (breaks apart) into hydrogen (H^+) and hydroxide (OH^-) ions
- $\text{H}_2\text{O} \leftrightarrow \text{H}^+ + \text{OH}^-$
- **Pure water** has equal concentrations of H^+ and OH^-
- **Acids** ($\text{pH} < 7$): high concentration of H^+ , taste sour or tart (citruses, gastric acid)
- **Bases** ($\text{pH} > 7$): High concentration of OH^- , taste bitter (bleach, soap); basic can also be referred to as “alkaline”



pH

- Used to rank solutions in terms of acidity or alkalinity (how basic it is)
- Measures concentration of H^+ ions
- MOST solutions fall on this scale but it is possible for solutions to be less than zero or more than 14
- Most cells exist within a pH of 7 and exposure to more acidic or basic conditions can cause denaturation of proteins





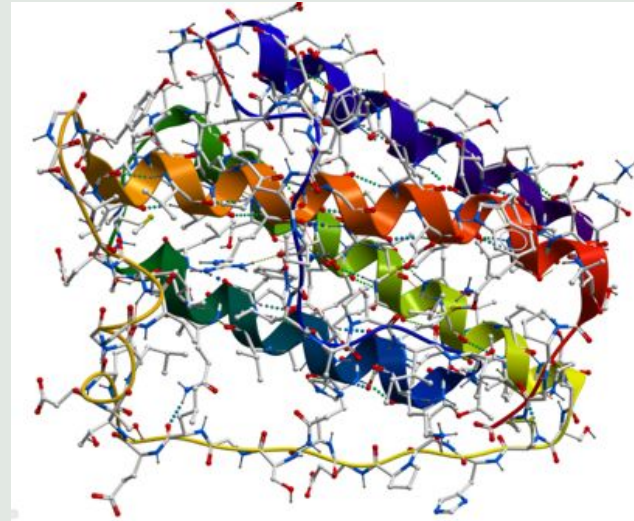
Organic Chemistry

Table of Contents :

1. Organic Compounds
 - a. Monomers vs. Polymers
 - b. Catabolic vs. Anabolic Reactions
 - i. Hydrolysis & Dehydration
2. Macromolecules
 - a. Carbohydrates
 - b. Lipids
 - c. Proteins
 - d. Nucleic Acids

Overview

- Macromolecules: large organic molecules (carbon-containing) that make up all living things
 - Carbohydrates, lipids, proteins, and nucleic acids
 - *We can't live without them!!*
 - *They run our body, hold information for how to run our body, and provide the energy to do so*
 - **EVERYTHING we will learn for the rest of the semester will relate back to macromolecules!!!**
 - Structurally, most macromolecules are polymers made out of monomers



Organic Compounds



organic – containing **carbon-carbon** and **carbon-hydrogen bonds** , usually with multiple carbons



Usually very large molecules



Usually contain C H, O, N, P, and sometimes S



Organic compounds make up and are responsible for all living processes!

Organic Compounds Cont.

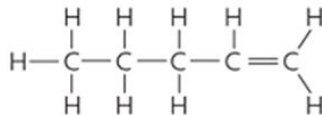
Carbon-based compounds typically come in 3 shapes:

□ **Straight Chain**

□ **Branched Chain**

□ **Ring**

Straight chain

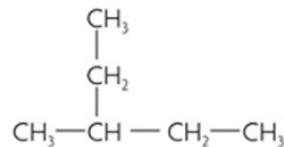


A simplified structure can also be shown as:



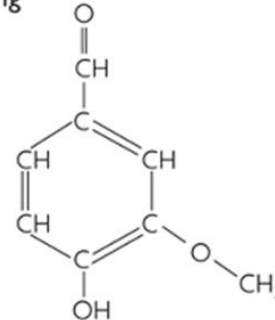
Pentene

Branched chain

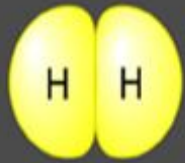


Hexane

Ring



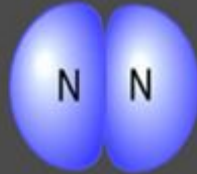
Vanillin



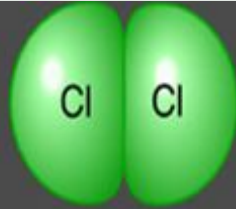
H_2
hydrogen



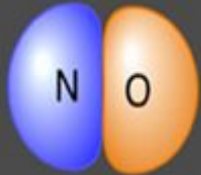
O_2
oxygen



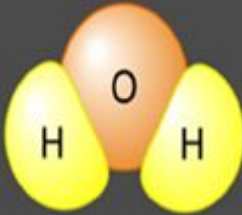
N_2
nitrogen



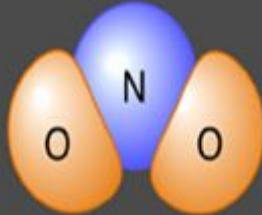
Cl_2
chlorine



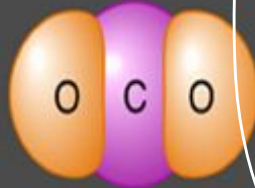
NO
nitrogen oxide



H_2O
water



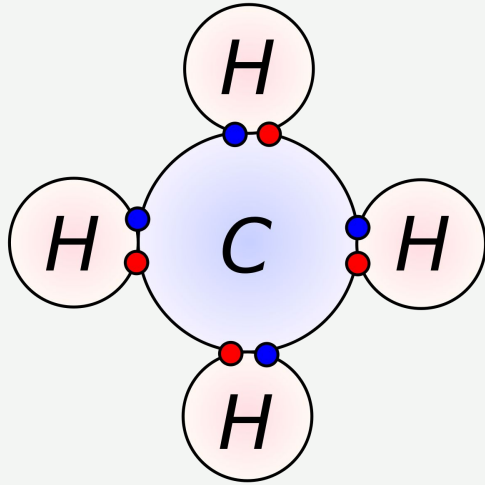
NO_2
nitrogen dioxide



CO_2
carbon dioxide

Inorganic compounds

What do inorganic compounds lack?



● Electron from hydrogen

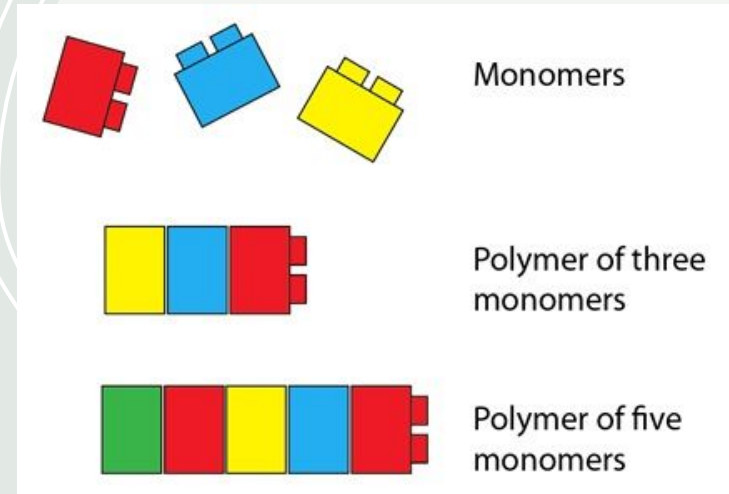
● Electron from carbon

Carbon Chemistry

- Carbon (C) has **4 valence electrons** → 4 electrons in its outermost shell
 - gives carbon **4 places** for covalent bonds to occur
- Allows for GREAT variation in size and structure of organisms

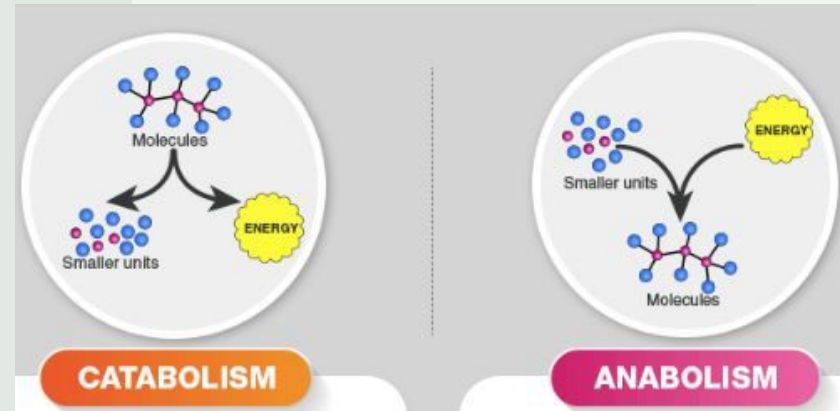
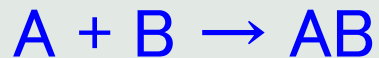
Monomers vs. Polymers

- many carbon molecules are made of many subunits bonded together
- monomers – small, basic subunits → “building blocks”
- polymers – larger, more complex structures made up of many monomers bonded together



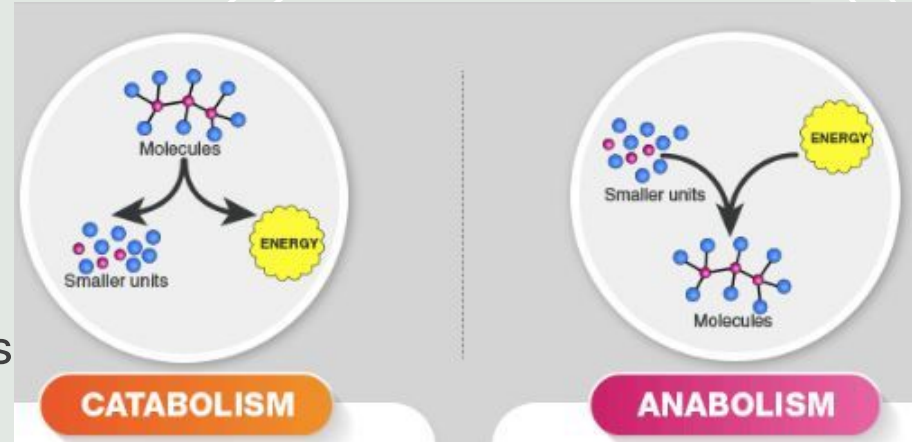
Anabolic Reactions

- anabolic reactions – processes that build up larger molecules from smaller units
- USE energy to synthesize large molecules



Catabolic Reactions

- catabolic reactions – processes that break reactants down
- RELEASE energy by making smaller molecules



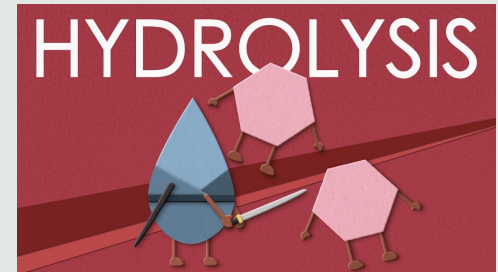
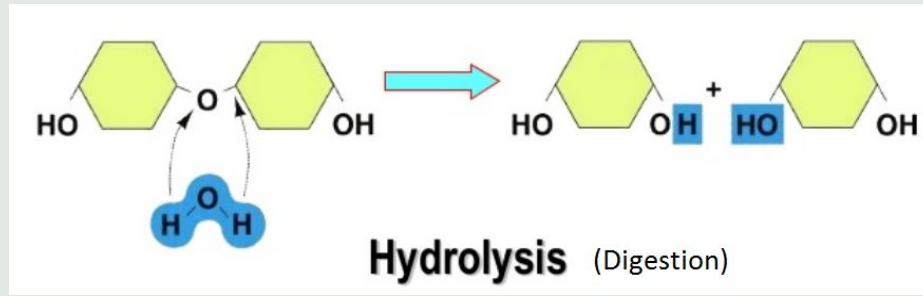
Building Polymers: Dehydration Synthesis

- Anabolic process that creates new bonds and larger molecules
- A type of **condensation** reaction (two molecules condensing into one)
- **Water is removed as a waste product** in EVERY place a new bond is formed
- An H is removed from one molecule and an OH is removed from the other

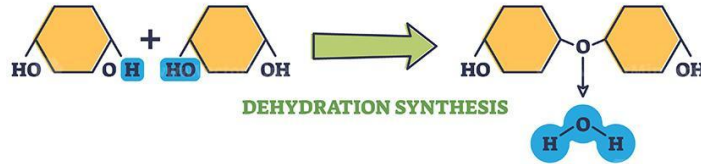


Breaking Polymers: Hydrolysis

- **Catabolic process** that breaks down bigger molecules into smaller ones
- **Water is added (“hydro-”)** as a reactant to break bonds of the polymer
- **“-lysis”** = to cut, break down
- A H will bond to one molecule while a OH will bond to the other



DEHYDRATION SYNTHESIS AND HYDROLYSIS



Metabolic pathways

Anabolic: Small molecules are assembled into large ones. *Energy is required.*



Catabolic: Large molecules are broken down into small ones. *Energy is released.*

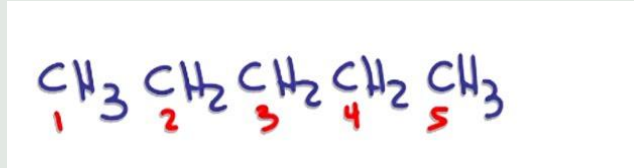


Carbon Skeletons

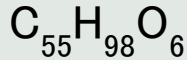
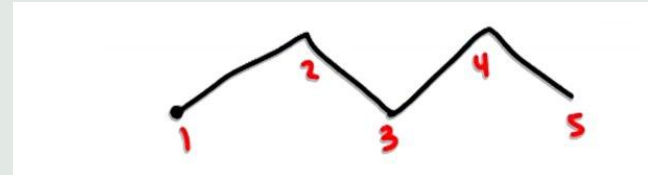
Remember : This is biology, not organic chemistry. You will not be asked to draw carbon skeletons, but many diagrams of macromolecules will be in skeleton form, and you should be able to recognize them.

- Skeleton = simplified zig-zag structure
- Carbons are found on “ends and bends”; each pointy end is a C atom
- Helpful to simplify and identify the structures of large molecules

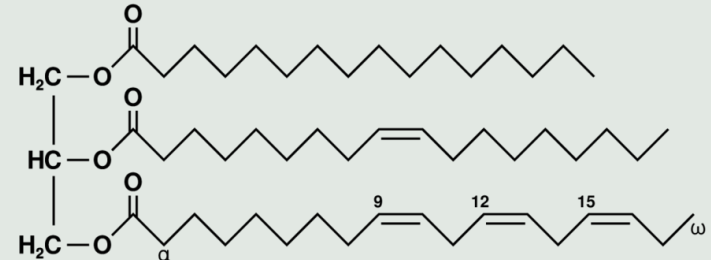
Formula

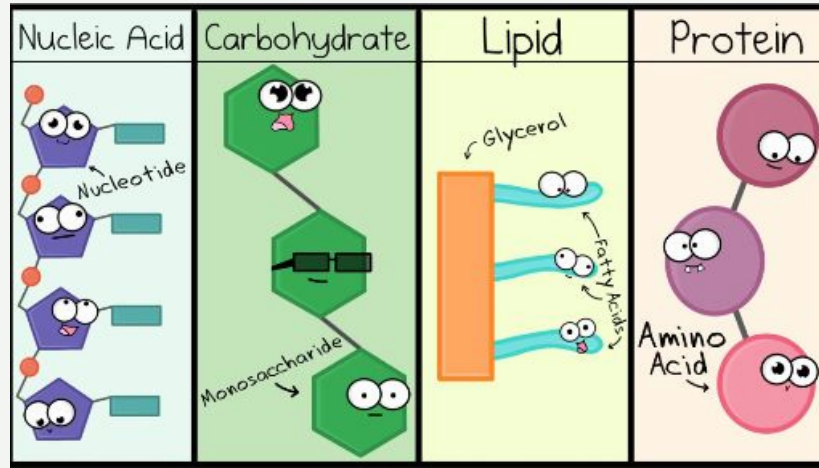


Skeleton



Formula for a triglyceride, a type of lipid

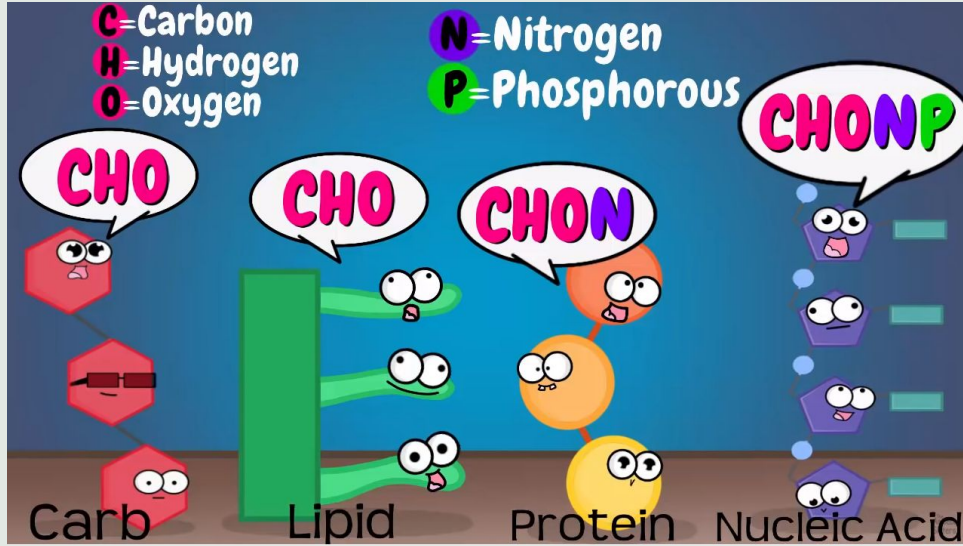




Macromolecules

Molecules of Life

- Also called biomolecules or polymers
- Made up of C, H, O, N, P and sometimes S (“CHONPS”)
- 4 types:
 1. Carbohydrates (“sugars”)
 2. Lipids (“fats”)
 3. Proteins
 4. Nucleic Acids (DNA and RNA)



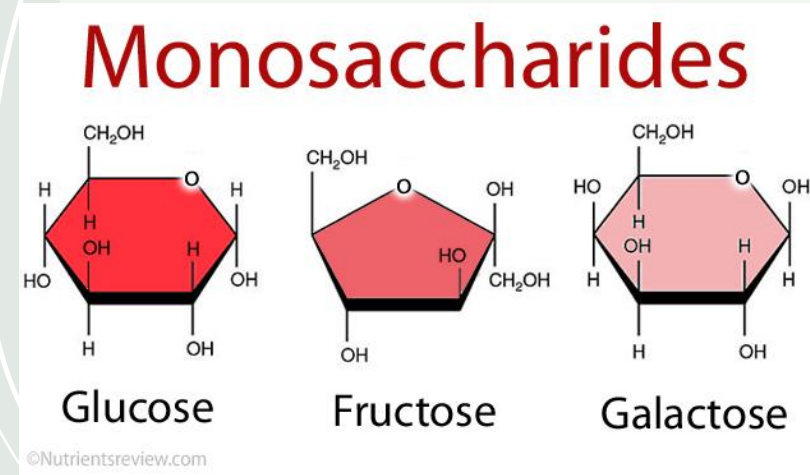
Summary of Elements

A circular collage of various healthy food items. In the center, there are several small white bowls containing different types of grains and legumes: quinoa, lentils, chickpeas, and rice. Surrounding these are fresh fruits like a whole orange, a banana, a bunch of purple grapes, and a red apple. There are also sliced carrots, a whole carrot, and some mushrooms. The background is a light, textured surface.

- **Elements:** C, H, O
- **Common identifiers:** Name usually ends in “-ose” (ex. glucose, sucrose, lactose, etc.), building blocks have 5- or 6-ring carbon ring
- **Building blocks:** monosaccharides
- **Main Function:** Used for short-term energy storage or quick energy
- **Bond between monomers:** glycosidic linkage

Carbs (cont): Monosaccharides

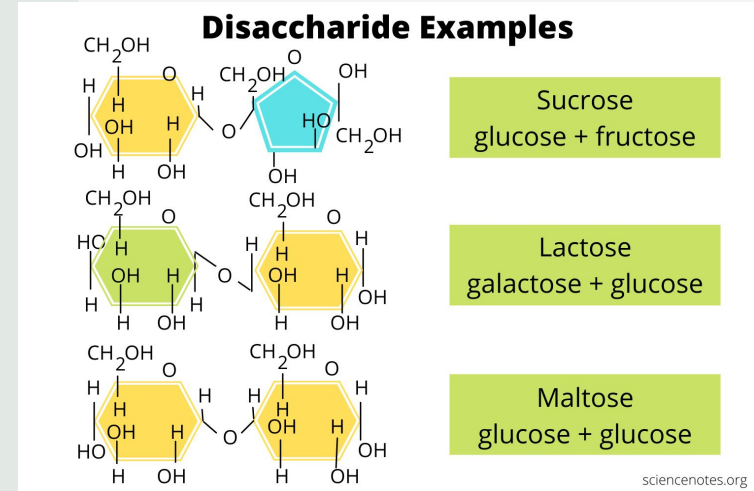
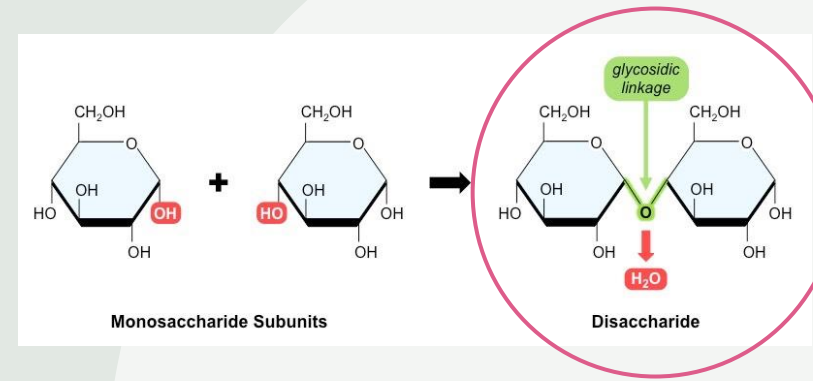
- monosaccharides – simple sugars that form **single carbon rings** with ratio of CH_2O
“mono-” = one “-saccharide” = sugar
- monomers of carbohydrates
- ex.) glucose, galactose, and fructose
- The human body can only metabolize sugars in monosaccharide form!



Carbs (cont.):

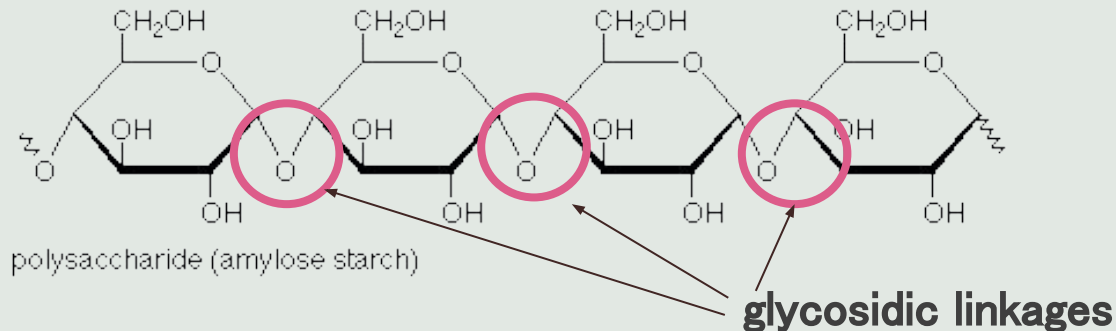
Disaccharides

- disaccharide – 2 monosaccharides bonded together with a **glycosidic linkage**
- ex.) sucrose (table sugar), lactose (milk sugar), maltose (milkshakes, beer, milk duds)



Carbs (cont.): Polysaccharides

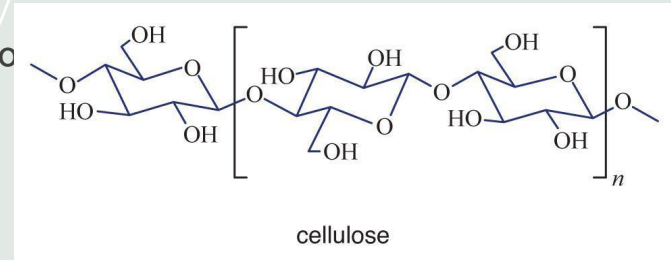
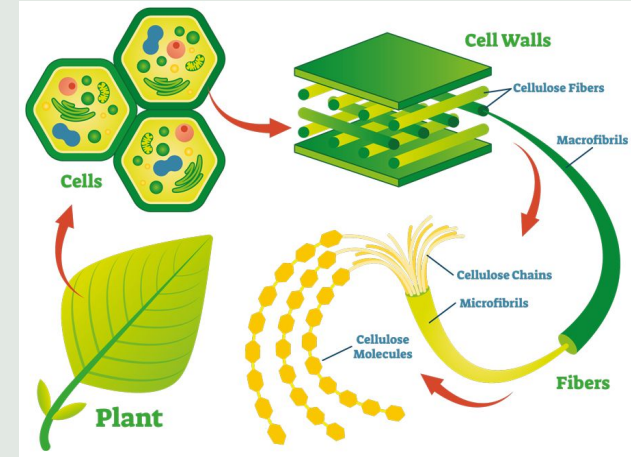
- **polysaccharides** – polymer made up of long chains of monosaccharides held together by **glycosidic linkages**
- ex.) **starch** in rice and pasta, **cellulose** in plants
- The body relies on **enzymes** to break polysaccharides down into monosaccharides



Polysaccharides in Plants

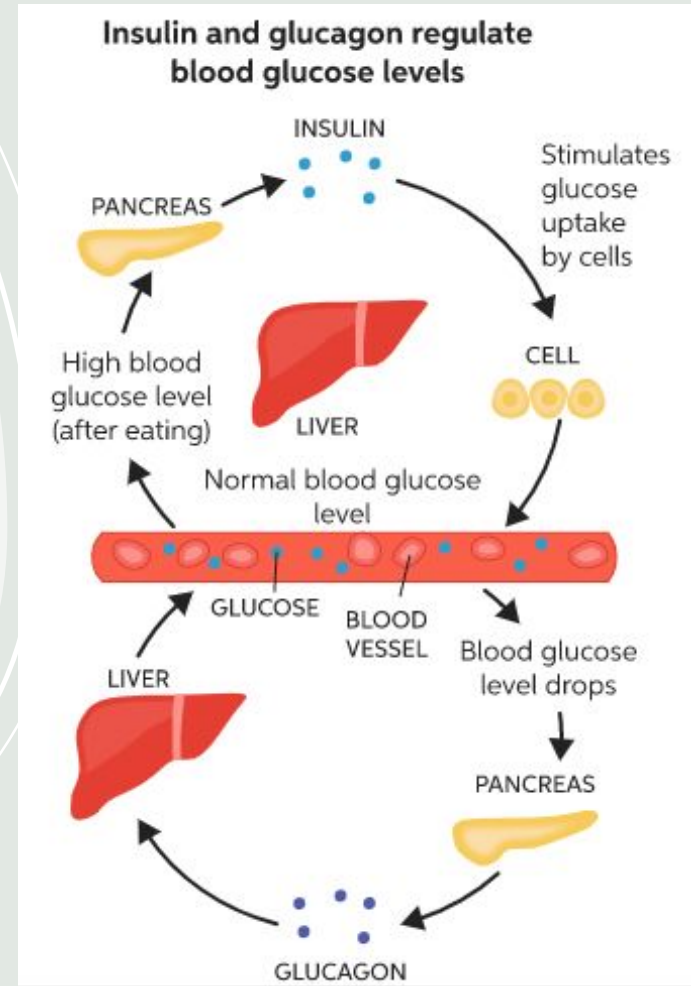
- **cellulose** – plant structural support
 - monomer = glucose
 - found in cell walls
 - fibrous and tough; not digestible by animals

(How can cows eat it, then???)
- **starch** – plant energy storage
 - monomer = glucose
 - stored in the roots of plants & tubers such as potato fruit, carrots, etc.
 - digestible with the help of enzymes (amylase)



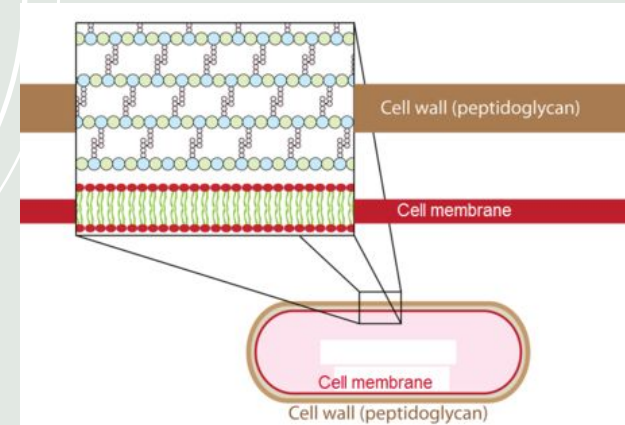
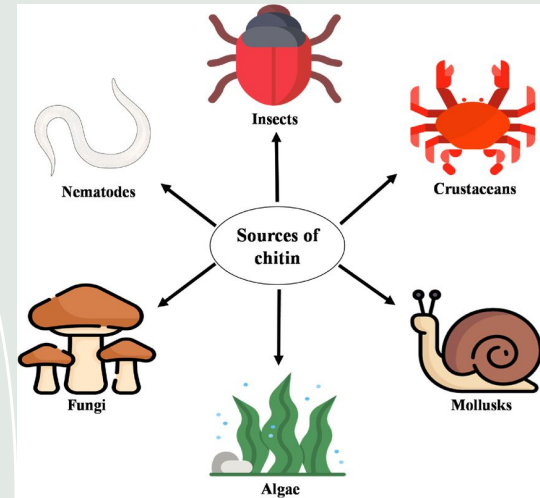
Polysaccharides in Animals

- glycogen – serves as **glucose storage** in **animals**
 - stored in **muscles and liver**
- provides short-term glucose storage between meals
 - glycogen reserves last about 24 hours when fasting, and only about an hour with vigorous exercise!
- body can break down glycogen to release glucose for energy



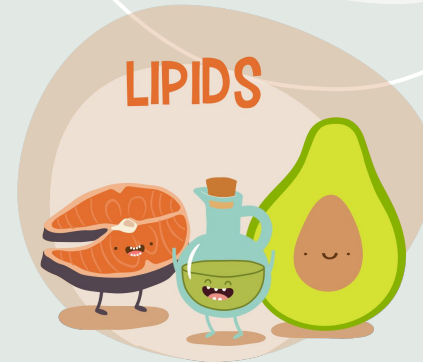
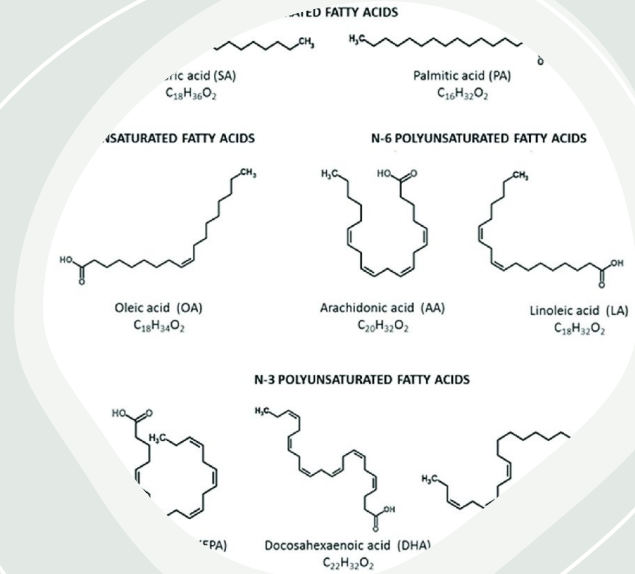
Polysaccharides in Other Organisms

- chitin – polysaccharide in the **cell walls** of **fungi** and in the exoskeletons of **insects** and **crustaceans**
- peptidoglycan – polysaccharide in the **cell walls** of **bacteria**
- give structure and support to these organisms



Lipids

- Elements: C, H, few O's
- Common identifiers: ALL are insoluble in water (**lipids are nonpolar**); some have distinct shapes
- Bonds between monomers: ester linkage
- Examples:
 - Fats (butter and lard)
 - Oils (Olive oil, peanut oil)
 - Triglycerides
 - Phospholipids
 - Steroids
 - Waxes



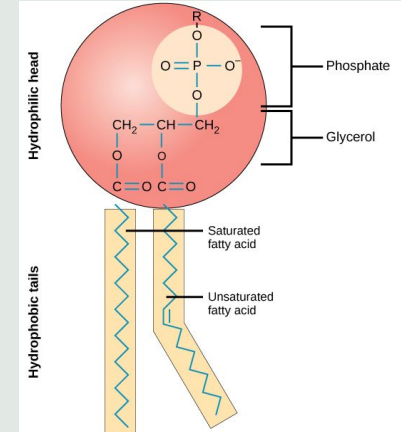
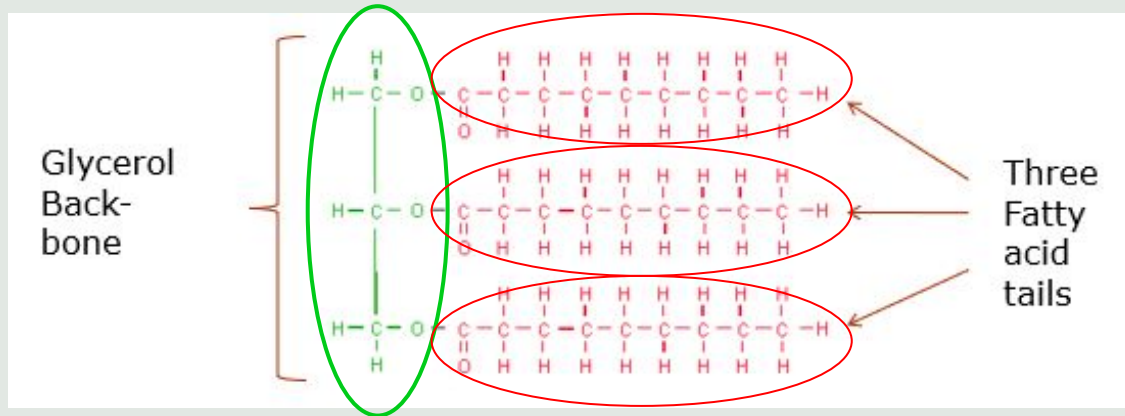
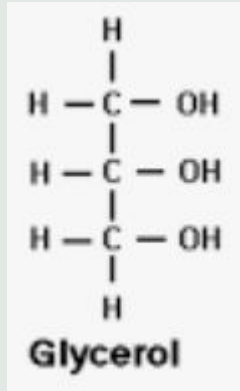
Lipids: Major Functions

- **LONG-TERM energy storage**
- Contain more C-H bonds than carbs, which can be broken for energy = **more energy per gram**
- Provides **insulation** (ex. blubber on sea animals, fat around important organs)
- Forms protective **water-repellent coatings**
 - wax and oil on wings, fur, leaves
- Forms **steroid hormones** (testosterone & estrogen)



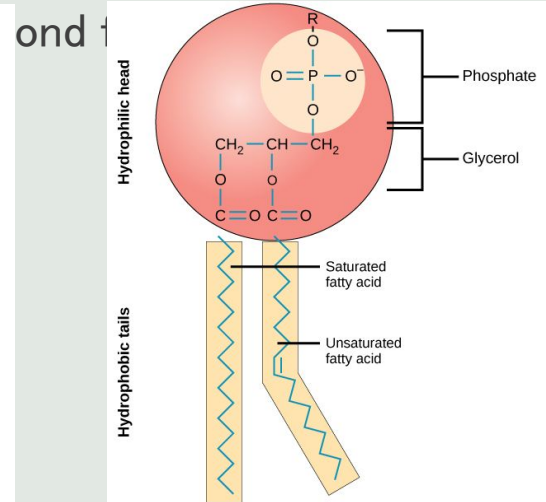
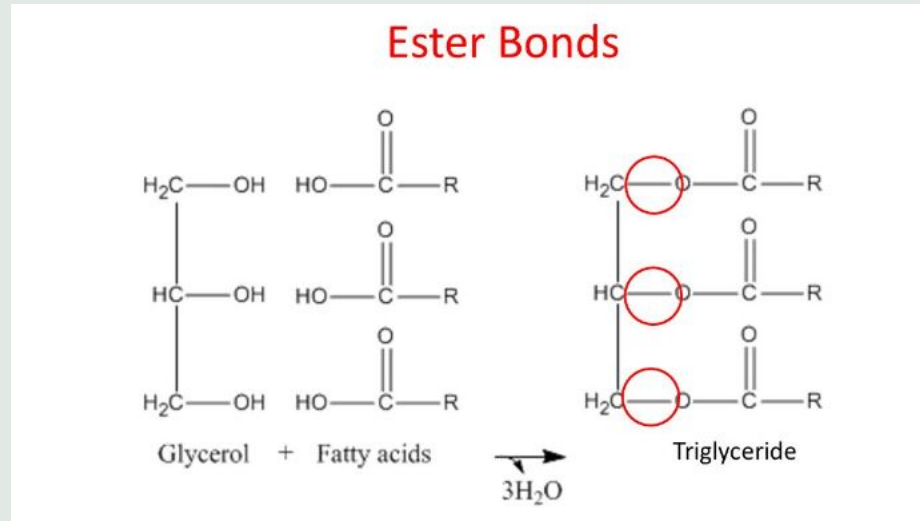
Lipids: Structure

- no traditional monomer: building blocks of **glycerol + fatty acids**
- no traditional polymer: can form large molecules such as triglycerides
- **glycerol** – molecule that can bond to 3 fatty acid tails to form a triglyceride; can also connect a phosphate and 2 fatty acid tails to form a phospholipid

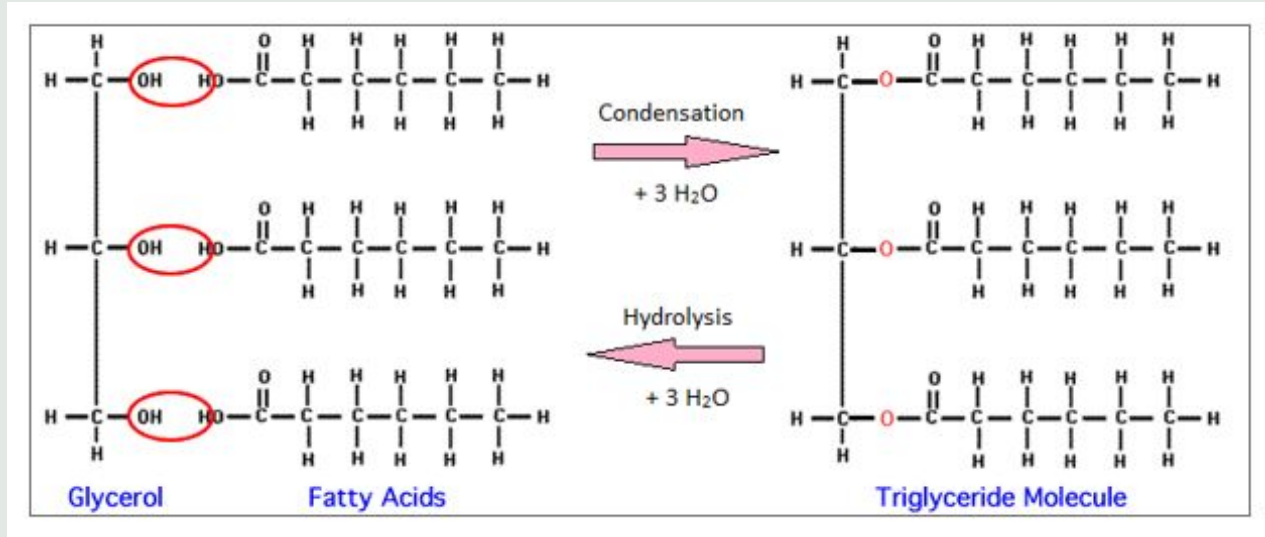


Lipids: Structure (cont.)

- glycerol and fatty acid tails are joined via **dehydration synthesis/condensation**
- ester linkage** – a covalent **bond** formed in **lipids** between glycerol and fatty



Typical Lipid Structure: A Triglyceride

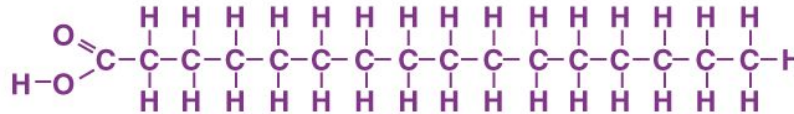


Note: This is a **saturated** triglyceride, because all three of its fatty acid tails are saturated (single C-C bonds only)

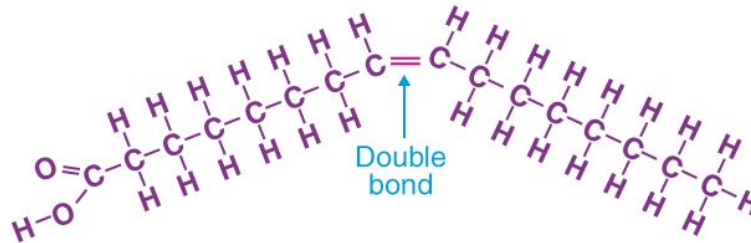
Two Different Types of Fatty Acids

- fatty acids – long chains of C atoms with H attached to them
- can be used for energy and for cellular structures

Saturated Fatty Acid



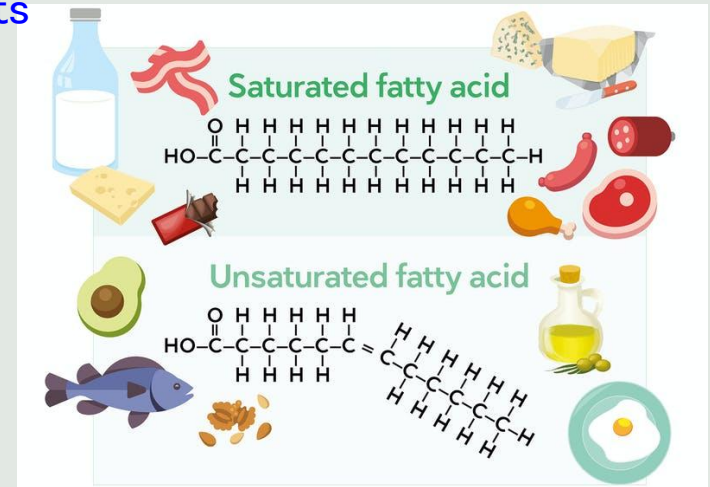
Unsaturated Fatty Acid



Saturated vs. Unsaturated Fatty Acids

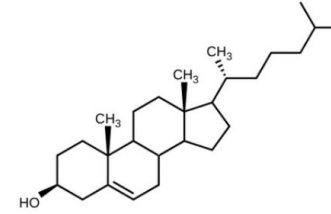
- saturated fatty acids – only contain **single bonds between C and C**; tend to be **solid** at room temp; “**bad fats**”
- unsaturated fatty acids – contain **double bonds between C and C**; tend to be **liquid** at room temp; “**good fats**”

Solid bacon
grease =
saturated fat

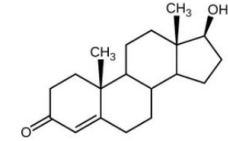


Steroids

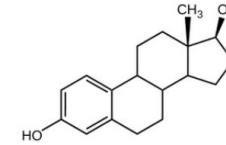
- steroids – lipids with a carbon skeleton consisting of **four fused rings**
- do not have glycosidic or ester bonds
- do not contain fat or glycerol
- classified as lipids due to hydrophobic nature
- ex.) testosterone, estrogen, cholesterol



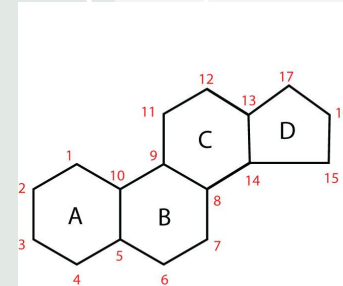
(a) Cholesterol



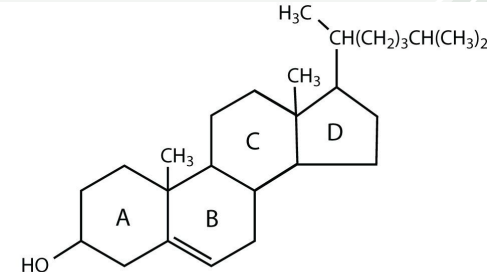
(b) Testosterone



(c) Estradiol

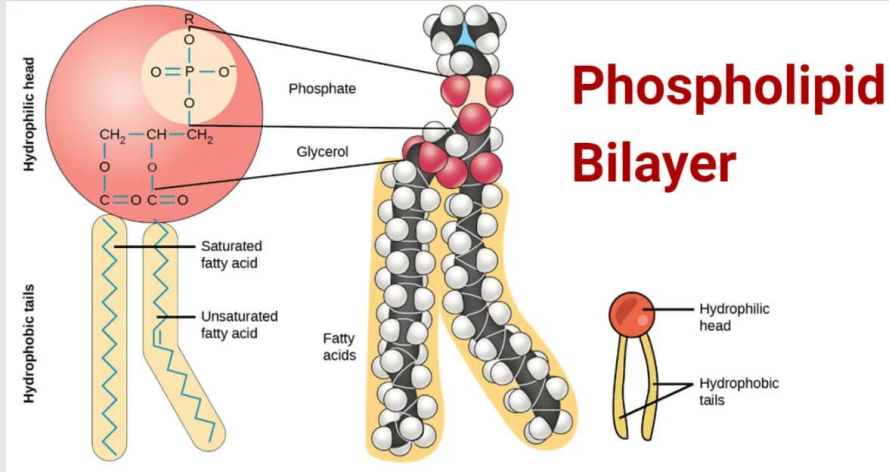
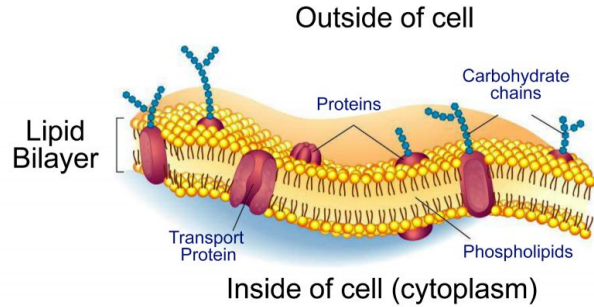


(a) Steroid skeleton



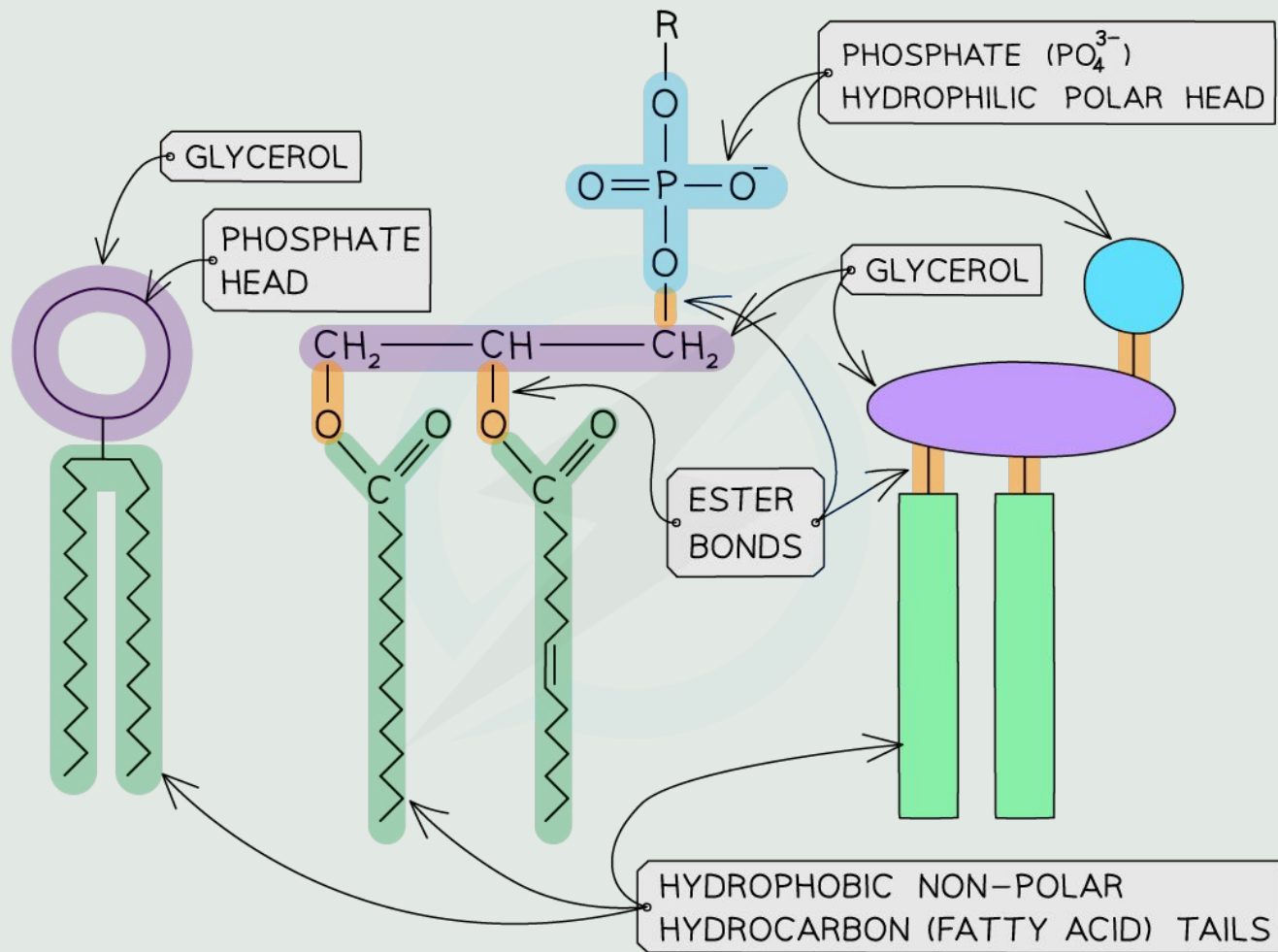
(b) Cholesterol

Structure of the Cell Membrane



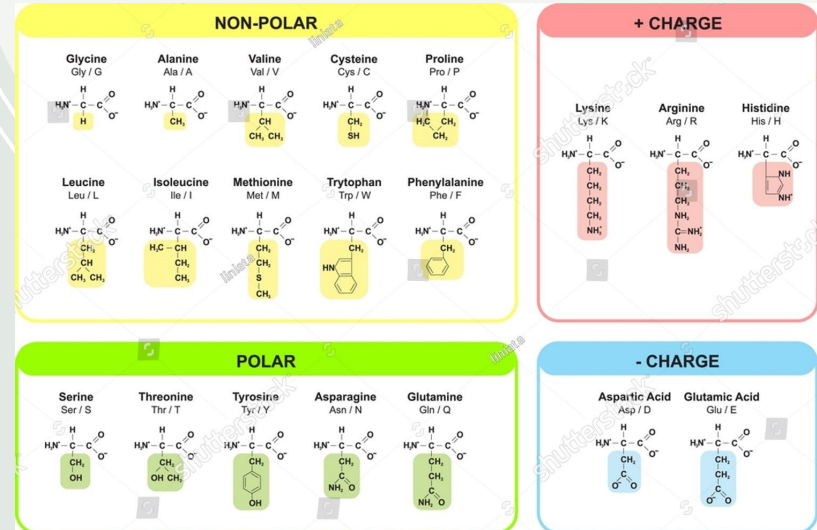
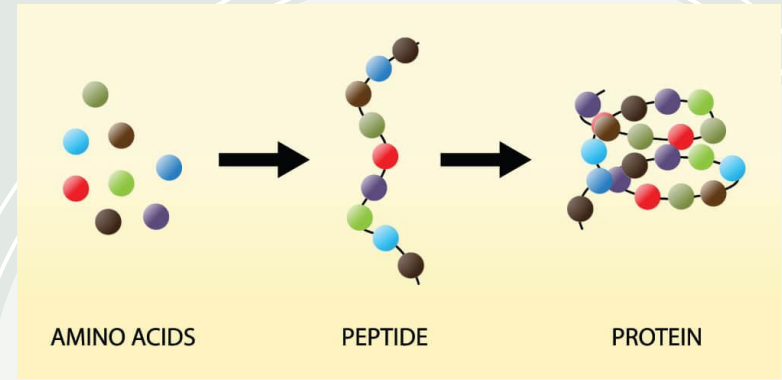
Phospholipids

- phospholipids – special lipid that makes up the bilayer of cell membranes
- Dual nature allows cell membrane to be *selectively permeable*
- 3 main parts:
 1. **phosphate group**
 2. **glycerol backbone**
 3. **two fatty acid tails** (one saturated straight, one unsaturated kinked)

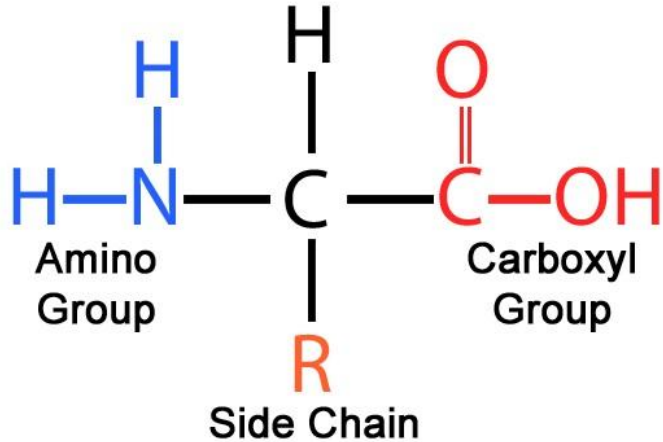


Proteins

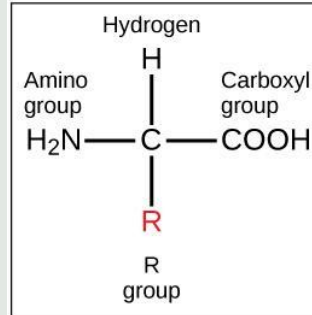
- **Elements** : C, H, O, N (and S) “CHONS”
- **Building blocks** : **amino acids (AAs)** ;
only **20 AAs** make up ALL life
- **Bond between monomers:** peptide
bonds between amino & carboxyl
groups
- Proteins (aka polypeptides) = chains of
many connected AAs



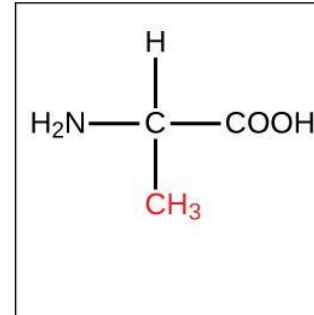
Amino Acid Structure (3 parts)



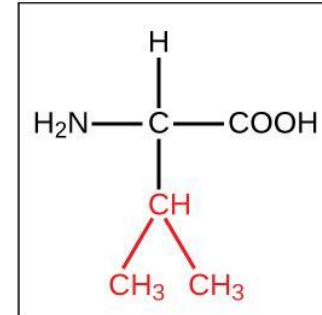
Fundamental structure



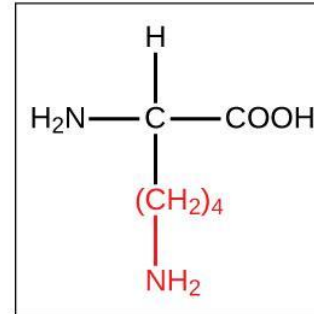
Alanine



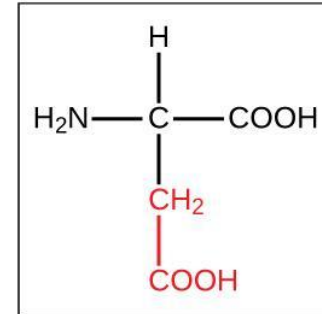
Valine



Lysine



Aspartic acid

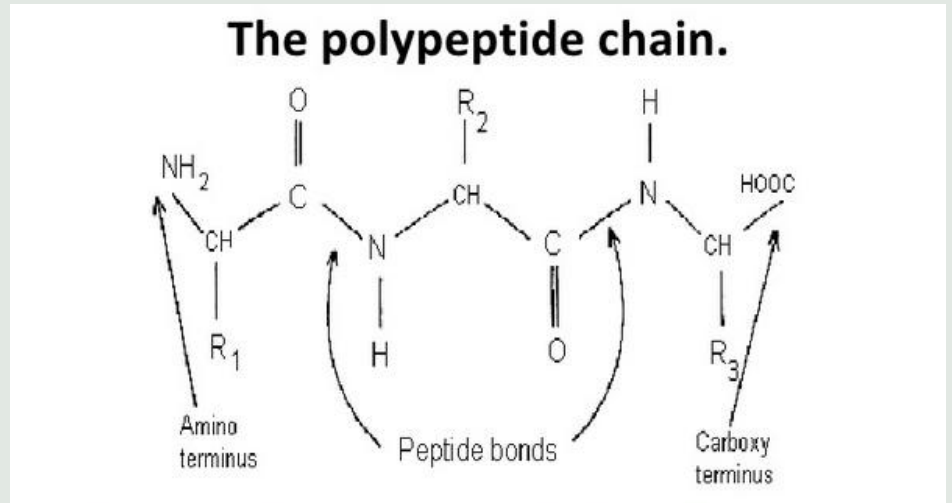
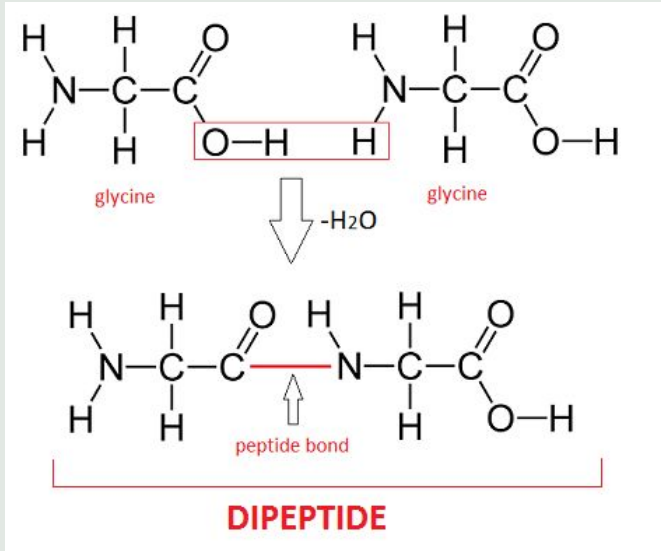


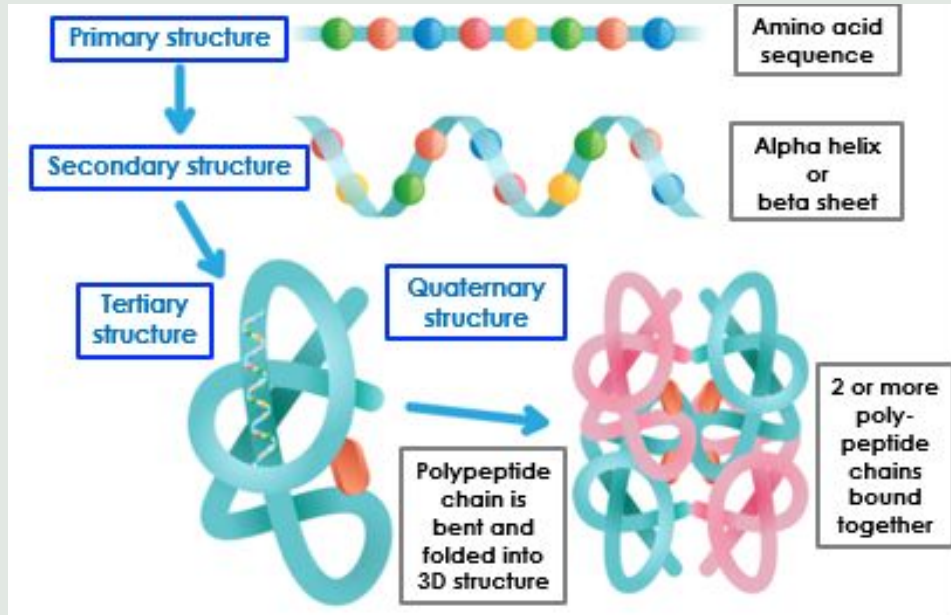
Role	Example	Function
Digestive Enzyme	Amylase, pepsin	Break down nutrients to be absorbed
Transport	Hemoglobin	Carry substances through body in blood
Structure	Actin, keratin	Build structures
Hormone Signaling	Insulin	Coordinate activity of body systems
Defense	Antibodies	Protect from pathogens
Contraction	Myosin	Carry out muscle contraction

Proteins do ALL THE THINGS

- no one “main function”
- build and repair tissue after damage (such as after exercise)
- **CORE IDEA: diverse shapes = diverse functions**

Protein Structure

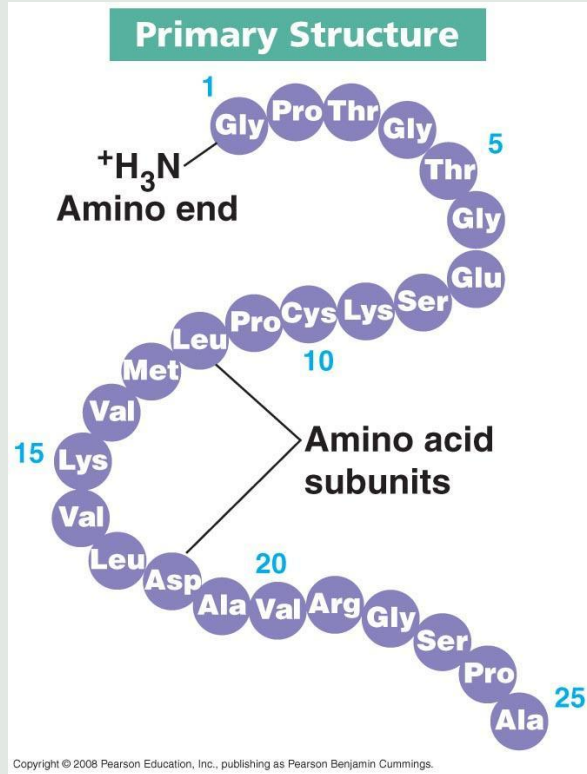




Protein Structure

- Proteins are the **most diverse** in structure and function
- **4 levels** to protein structure
- **CORE IDEA:**

structure = function □ protein shape determines its function

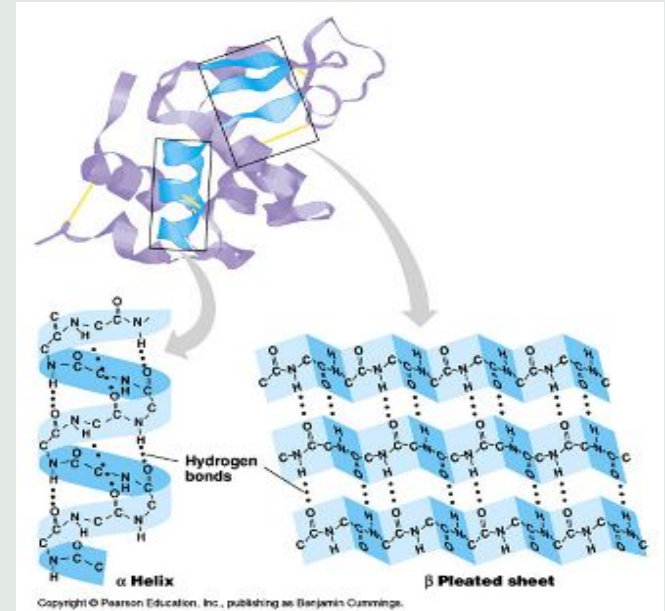


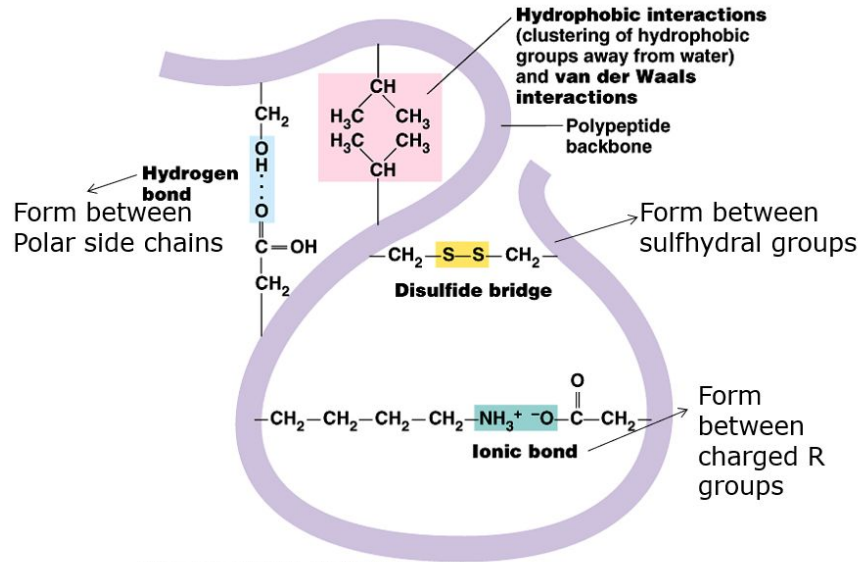
Primary Protein Structure

- primary structure – the **sequence of AAs** that makes up a protein, held together by **peptide bonds**
- “backbone” of protein structure
- sequence is **determined by your DNA sequence**

Secondary Protein Structure

- secondary structure – folded structures that form due to interactions between atoms of the backbone
- 2 types: **alpha (α) helix** and **beta (β) pleated sheet** held in shape by hydrogen bonds



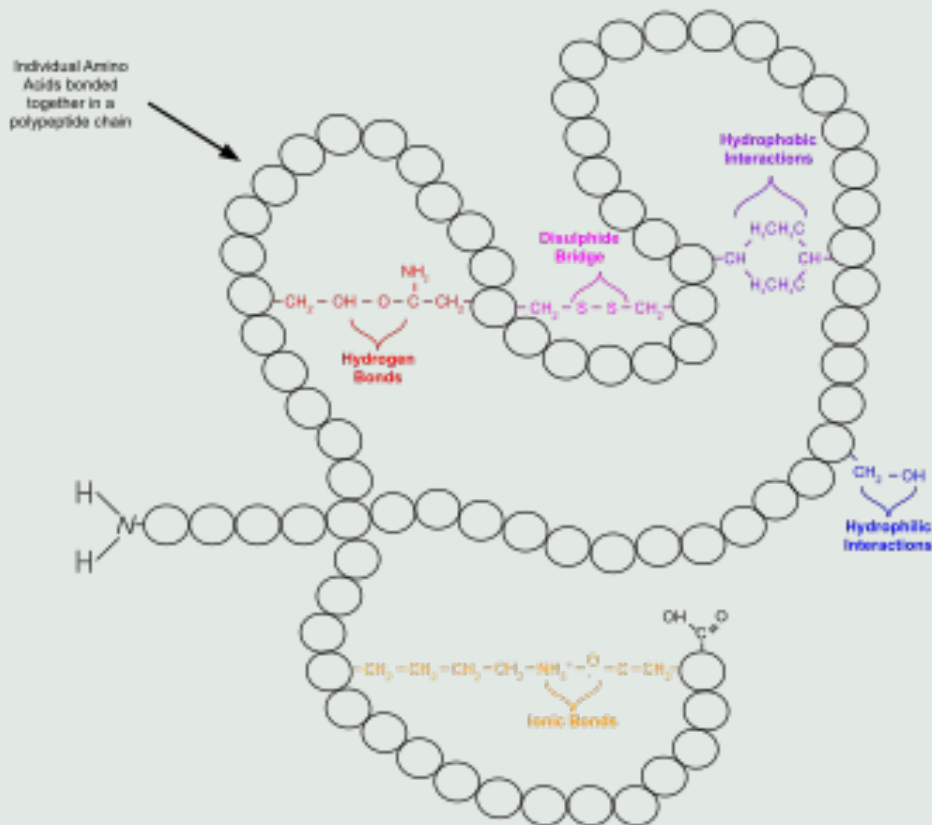


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Tertiary Protein Structure

- tertiary structure – overall 3D structure of a polypeptide
- due to **interactions between R groups (side chains)**

R Groups



Bond Types

Hydrophobic Interactions:

These amino acids orient themselves towards the center of the polypeptide to avoid the water

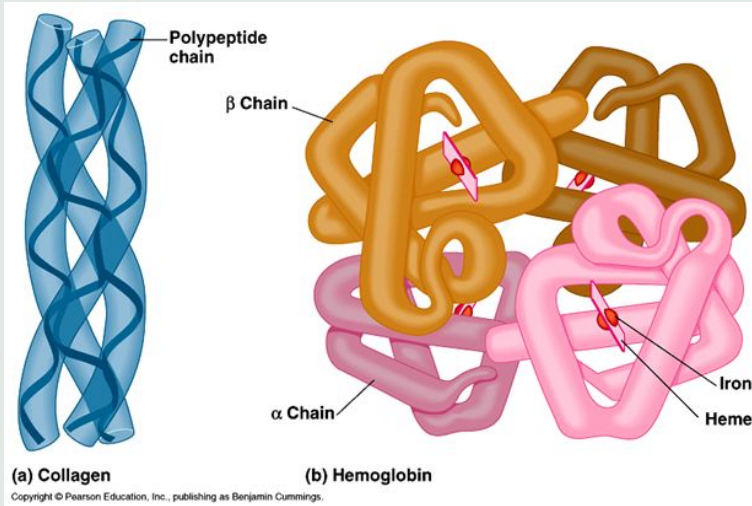
Disulphide Bridge: The amino acid cysteine forms a bond with another cysteine through its R group

Hydrogen Bonds: Polar "R" groups on the amino acids form bonds with other Polar R groups

Hydrophilic Interactions:

These amino acids orient themselves outward to be close to the water

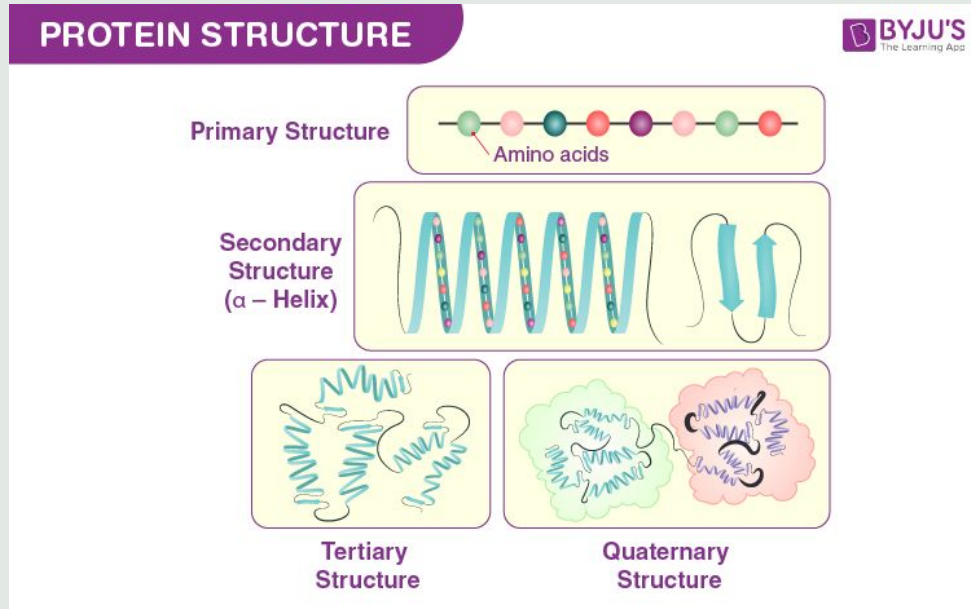
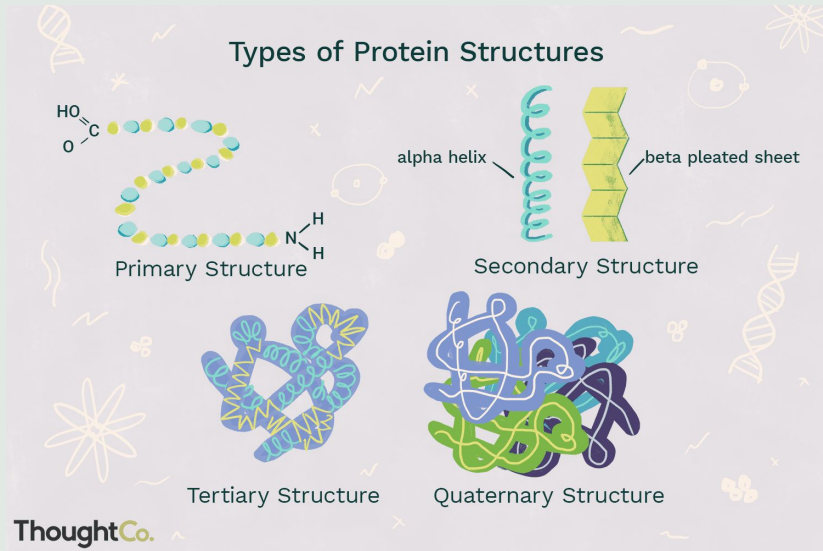
Ionic Bonds: Positively charged R groups bond together



Quaternary Protein Structure

- quaternary structure – structure in proteins that results from interactions between and among *several* polypeptide chains
- only possible when a protein has multiple subunits (ex. hemoglobin has 4 subunits)

Summary of Protein Structures



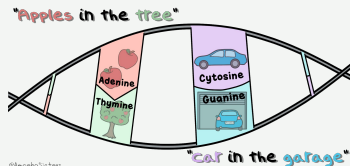
Nucleic Acids

- Elements : CHONP
- Building blocks : nucleotides
- Polymers (2) : DNA and RNA
- Bonds : **phosphodiester bonds** in the backbone, **hydrogen bonds** between the bases
- Function : stores all **genetic info** & instructions for making **proteins**; molecule of heredity

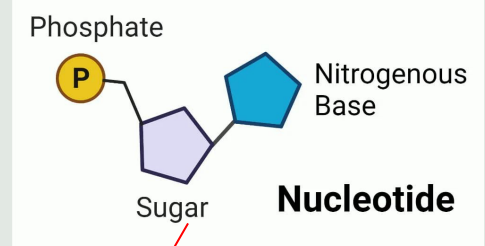
5 Nitrogenous Bases & Base Pairing :

Adenine & Thymine or Uracil (DNA or RNA)

Guanine & Cytosine



Nucleotide (3 components: phosphate, sugar, and base)



DNA Strand
(comprised of many nucleotides)

